



MARGAM: MATERNAL ASSISTANCE FOR RURAL GUIDANCE AND MONITORING

A MINI PROJECT REPORT

Submitted by

MELBIN K MADHU

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Under the Guidance of

Ms. KAVITHA S

(Assistant Professor, Department of Computer Applications)



**DEPARTMENT OF COMPUTER APPLICATIONS
MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR**

(Affiliated to APJ Abdul Kalam Technological University)

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VISION

To become a centre of excellence in computer applications, competent in the global ecosystem with technical knowledge, innovation with a sense of social commitment.

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- To serve with state of the art education, foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

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Program Specific Outcomes

- **PSO I:** Apply advanced technologies through innovations to enhance the efficiency of design and development.
- **PSO II:** Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	Applied knowledge of database design, web architecture and software engineering principles to build MARGAM.
2	PO2	Analysed the fragmented, paper-based maternal and child health tracking process in rural panchayats to define system requirements.
3	PO3	Designed a role-based Django solution covering Government, ASHA Worker, Doctor and Mother workflows.
4	PO5	Used Django 5, Bootstrap 5.3.3, Chart.js, SQLite3, Visual Studio Code and GitHub Copilot as modern development tools.
5	PO9	Project was independently designed, implemented and tested under the periodic guidance of the project guide.
6	PO12	Involved self-directed learning of Django signals, role-based access control and data visualisation with Chart.js.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (Foundation Knowledge): Apply knowledge of programming logic, database concepts and coding fundamentals for solution architecture and problem solving.

PO2 (Problem Analysis): Identify, review, formulate and analyze problems in maternal and child healthcare coordination with focus on user requirements.

PO3 (Development of Solutions): Design and develop an integrated web-based healthcare solution for maternal registration, vaccination tracking, appointment management and healthcare coordination.

PO4 (Modern Tool Usage): Select and apply modern computational tools and technologies such as Python, Django, Bootstrap, SQLite3 and JavaScript for system development.

PO5 (Individual and Teamwork): Function and communicate effectively as an individual or team member during the planning, development and testing of the project.

PO6 (Project Management and Finance): Apply project management principles such as planning, scheduling, resource management and work breakdown structure.

PO7 (Ethics): Follow professional ethics and software security practices to protect maternal, child, appointment and communication information from unauthorized access.

PO8 (Life-long learning): Develop change management skills and continuously learn and adapt to contemporary technologies and methods for improving digital healthcare systems.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO1 – Apply advanced technologies through innovations to enhance the efficiency of design development	Django, Bootstrap 5, SQLite3, role-based authentication, automated vaccination scheduling, notifications, and dashboards were applied to design an integrated maternal and child healthcare platform.
2	PSO2 – Apply the principles of computing to analyze, design and implement sustainable solutions for real-world challenges	Computing principles of database design, web development, software architecture, access control, and workflow automation were applied to build a sustainable solution for maternal and child healthcare coordination.

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 3	Good Health and Well-Being	The platform supports maternal and child healthcare through pregnancy information, vaccination tracking, appointments, notifications, and follow-up activities.
SDG 5	Gender Equality	The Mother portal gives women direct access to maternal health information, vaccination schedules, appointments, and communication with ASHA Workers.
SDG 9	Industry, Innovation and Infrastructure	Django 5, Bootstrap 5, and SQLite3 are used to create an integrated digital healthcare platform with role-based portals and automated vaccination generation.
SDG 10	Reduced Inequalities	The platform connects Government officials, ASHA Workers, Doctors, and Mothers through one system, helping improve access to organized maternal and child healthcare services.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 1 – No Poverty: End poverty in all its forms everywhere.

SDG 2 – Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality: Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

SDG 8 – Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

SDG 9 – Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation.

SDG 10 – Reduced Inequalities: Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient and sustainable.

SDG 12 – Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action: Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water: Conserve and sustainably use the oceans, seas and marine resources.

SDG 15 – Life on Land: Protect, restore and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification and halt biodiversity loss.

SDG 16 – Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions.

SDG 17 – Partnerships for the Goals: Strengthen the means of implementation and revitalise the global partnership for sustainable development.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
OCTOBER 2026



DECLARATION

I hereby certify that the work presented in the project entitled “MARGAM: Maternal Assistance for Rural Guidance And Monitoring” submitted to the DEPARTMENT OF COMPUTER APPLICATIONS is an authentic record of my own work carried out under the supervision of Ms. Kavitha S, Assistant Professor. This work has not been submitted to any other institution or university for the award of any other degree. The report has been checked for plagiarism by the college and the similarity index is within the permissible limit set by the college.

Name & Signature of Student

MELBIN K MADHU

Date:

Place:

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
OCTOBER 2026



CERTIFICATE

*Certified that the report entitled “**MARGAM: Maternal Assistance for Rural Guidance And Monitoring**” submitted by **MELBIN K MADHU (MLM25MCA-2040)** to the **APJ Abdul Kalam Technological University** in partial fulfillment of the requirements for the award of **the Degree of Master of Computer Applications** is a bonafide record of the **MINI PROJECT** work carried out by the student under our guidance and supervision during the academic year 2025-27.*

PROJECT GUIDE

Ms. Kavitha S

Assistant Professor

Department of Computer Applications

PROJECT COORDINATOR

Ms. Athira R Kurup

Assistant Professor

Department of Computer Applications

HEAD OF THE DEPARTMENT

Ms. Divya S B

Associate Professor

Department of Computer Applications

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic “***MARGAM: Maternal Assistance for Rural Guidance And Monitoring***”.

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MELBIN K MADHU

MLM25MCA-2040

ABSTRACT

MARGAM (Maternal Assistance for Rural Guidance and Monitoring) is a role-based web application designed to improve coordination and information management for maternal and child healthcare services. The system brings together Government Officials, ASHA Workers, Doctors and Mothers through a centralized platform with role-specific dashboards and controlled access. It supports authentication, maternal and baby registration, baby growth monitoring, vaccination schedule management, doctor appointment booking, health information, notifications and communication between mothers and their assigned ASHA Workers.

Government users can manage doctors and ASHA Workers, assign mothers, maintain vaccination schedule templates, manage health-related content, issue notifications and view reports. ASHA Workers can manage assigned mothers and babies, record growth information, update eligible vaccination records and communicate with mothers. Doctors can configure availability, handle appointments, view authorized patient information and record clinical updates. Mothers can manage their own information and baby records, monitor growth and vaccination information, request appointments and communicate with assigned ASHA Workers.

A key workflow automation is the generation of vaccination records when a baby is registered, based on active vaccination schedule templates maintained by authorized government users. The appointment module validates doctor availability and capacity before creating bookings and supports controlled confirmation, rejection and completion workflows. The application was implemented using Python and Django with SQLite, HTML5, CSS3, Bootstrap 5 and JavaScript. Security hardening included relationship-aware object authorization, CSRF protection, POST-only state-changing actions, password-change support and production-safe host configuration.

The final application was verified through a comprehensive Django test suite containing 43 automated tests. The final full-suite execution completed with 43 tests passed and 0 failures, while the Django system check reported no issues.

Keywords: *maternal health, child health, web application, Django, ASHA Worker, vaccination management, appointment management, role-based access control, notifications, healthcare information system.*

Mapping with Sustainable Development Goals (Mention the Goal)	SDG 3 – Good Health and Well-being SDG 5 - Gender Equality SDG 10 – Reduced Inequalities SDG 16 - Peace, Justice, and Strong Institutions
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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ASHA	- Accredited Social Health Activist
CSRF	- Cross-Site Request Forgery
CSS	- Cascading Style Sheets
DFD	- Data Flow Diagram
EMR	- Electronic Medical Record
HTML	- HyperText Markup Language
MARGAM	- Maternal Assistance for Rural Guidance And Monitoring
MCH	- Maternal and Child Health
ORM	- Object-Relational Mapping
PO	- Program Outcome
PSO	- Program Specific Outcome
SDG	- Sustainable Development Goal
SMS	- Short Message Service

CHAPTER 1

INTRODUCTION

1.1 Background

Maternal and child healthcare management has moved steadily from paper-based records and manual coordination towards digital, technology-enabled healthcare services. Government health departments, ASHA Workers, Doctors, and Mothers need to manage information across different stages of maternal and child care, and it is increasingly important for registration, vaccination, appointments, and follow-up activities to be organized through a centralized platform. Among these activities, continuous healthcare coordination is the most demanding. It must provide not only accurate maternal and child information, but also support vaccination tracking, appointment management, timely notifications, healthcare follow-up, and effective coordination between the different users involved in the process.

A meaningful maternal and child healthcare platform therefore requires several capabilities working together: secure user authentication so that each authorized user can access the appropriate services; a centralized database in which maternal, child, vaccination, appointment, and healthcare information can be maintained; a system for generating and tracking vaccination schedules; a way to manage appointments and healthcare activities; a dashboard through which users can monitor relevant information; and a structured method of maintaining records and providing notifications. In practice these capabilities are often handled through separate registers, documents, or disconnected systems. Government officials maintain administrative information, ASHA Workers coordinate with Mothers, Doctors manage healthcare-related activities, and Mothers depend on manual communication to obtain updates about vaccinations and appointments.

Recent web technologies make it possible to integrate these services into a single browser-based platform. Django provides a structured framework for secure web development, while Bootstrap enables responsive interfaces. SQLite3 stores maternal, child, vaccination, appointment, and user data. Role-based access allows Government officials, ASHA Workers, Doctors, and Mothers to use relevant functions. Automated vaccination schedules, appointments, notifications, and centralized records improve healthcare coordination and form the technical foundation of MARGAM.

1.2 Motivation

The motivation for this project comes from the everyday difficulties faced by mothers, ASHA Workers, Doctors, and Government officials in coordinating maternal and child healthcare services. Mothers often depend on manual communication to obtain information about vaccinations, appointments, and healthcare activities, while ASHA Workers have to coordinate and maintain records for multiple mothers and children. Government officials, on the other hand, need to manage healthcare-related information and monitor activities systematically, while Doctors require organized access to relevant maternal and child health details.

Existing healthcare management practices address parts of this problem. Paper-based records can store essential information but make updating, tracking, and retrieving records difficult; manual communication can provide information but often lacks a centralized record of activities; and separate digital tools may manage individual functions without providing an integrated platform for all stakeholders. These limitations can lead to difficulties in vaccination tracking, appointment coordination, record management, and timely follow-up. There is thus a clear opportunity for an integrated web-based platform that brings these services together, improves coordination among healthcare stakeholders, and provides Mothers with convenient access to their maternal and child healthcare information.

The project also offered the developer a valuable opportunity to apply a broad set of modern web technologies, including component-based user interfaces, role-based authentication, database-driven web applications, automated vaccination scheduling, notification management, and dashboard-based information systems, to a realistic maternal and child healthcare problem with immediate practical relevance.

1.3 Scope & Objectives

The scope of the MARGAM system includes secure authentication and role-specific access for four user groups. It maintains maternal profiles and ASHA assignments, supports baby registration with automatic vaccination-record generation, and stores baby growth measurements with visualization. The system also provides information on doctor availability and allows mothers to submit appointment requests.

The system supports controlled doctor workflows, including appointment confirmation, rejection, and completion. It enables authorized vaccination updates and provides government dashboards for personnel management, vaccination schedule management, notifications, and report generation. It also offers health tips, food-related information, and disease-related information to support maternal and child healthcare.

Additionally, MARGAM facilitates one-to-one communication between mothers and ASHA workers. The system incorporates essential security measures, including object-level authorization, CSRF protection, and POST-only state-changing operations, to ensure secure and controlled access to sensitive healthcare information.

1.4 Problem Statement

The core problem addressed by MARGAM is the absence of a single, role-aware system through which maternal and child health information, vaccination activities, appointments, follow-up activities and communication can be effectively coordinated. Manual or disconnected processes can lead to duplicated data entry, delayed updates, missing information and difficulty in maintaining accurate health records. Such limitations can reduce the efficiency of healthcare workers and make it difficult to provide timely support to mothers and babies.

From the application-development perspective, the problem has two major dimensions. The first is coordination, where information related to a mother, baby, vaccination, appointment and follow-up activity should remain connected rather than being maintained as unrelated records. A centralized system is required to maintain these relationships and provide relevant information to the appropriate users. The second dimension is controlled visibility, where the same centralized database must prevent users from viewing or modifying information beyond their assigned responsibilities.

MARGAM addresses these challenges by providing a structured web-based platform built using a role-based Django architecture. The system connects maternal records, baby details, vaccination schedules, growth information, appointments and communication within a unified environment while applying appropriate access restrictions for government officials, ASHA workers, doctors and mothers. This approach improves information organization, reduces unnecessary duplication, supports timely updates and helps ensure that sensitive healthcare information is accessed and modified only by authorized users.

CHAPTER 2

REQUIREMENT AND ANALYSIS

2.1 Existing System and Problem Statement

The existing workflow for maternal and child healthcare management is fragmented across paper-based records, phone calls and separate administrative activities. Such disconnected methods make it difficult to maintain a single, current and organized record of mothers, babies, vaccinations, appointments and follow-up activities. ASHA Workers may have to manage information related to multiple mothers and babies while also coordinating healthcare activities, whereas government users may not have a consolidated view of personnel, vaccination schedules, appointments and other operational information. These limitations can result in duplicated data entry, delayed updates, difficulty in tracking healthcare activities and reduced visibility of important information. Maintaining records manually can also make it difficult to ensure that information remains consistent across different users and activities.

Another important limitation of the existing approach is the difficulty in controlling access to healthcare information. Maternal and child health records contain sensitive information, and different stakeholders require different levels of access according to their responsibilities. A government official, ASHA Worker, doctor and mother should not have unrestricted access to the same information. When information is maintained through disconnected or manual processes, enforcing such boundaries becomes difficult and may result in unauthorized access or modification of records. There is therefore a need for a centralized system that can connect maternal profiles, baby records, vaccination activities, growth information, appointments and communication while maintaining appropriate access restrictions.

The major problem identified is the absence of a unified and role-aware platform that can coordinate these activities while maintaining proper information boundaries. A suitable system should not only store healthcare information but should also connect related records and provide controlled workflows for different stakeholders. MARGAM is proposed to address these limitations by providing a centralized web-based platform with role-based access for Government Officials, ASHA Workers, Doctors and Mothers. The system combines healthcare record management, vaccination tracking,

growth monitoring, appointment management, notifications and communication within a single environment, thereby improving organization, accessibility and control of maternal and child healthcare information.

2.2 Solution Overview

The proposed solution to address the identified healthcare management challenges is the development of the MARGAM web application. The application is designed as a centralized maternal and child healthcare management system that connects mothers, babies, ASHA Workers, doctors and government authorities through a common platform. The system provides an organized environment for maintaining healthcare records, monitoring vaccinations and growth, managing appointments, and supporting communication between stakeholders involved in maternal and child care.

The MARGAM application allows users to register and log into the system using secure authentication. After successful authentication, users are directed to role-based dashboards. Government Officials can manage system resources, personnel, vaccination schedules, notifications and reports. ASHA Workers can manage assigned mothers and babies and communicate with them. Doctors can manage availability and handle appointment requests, while Mothers can manage their information and baby-related records within their permissions. This role-specific structure ensures that each user can access the functions required for their responsibilities.

Another important feature is the connection between healthcare records and automated workflows. When a baby is registered, applicable vaccination records can be generated using active vaccination schedule templates, reducing manual entry and maintaining a structured vaccination history. Baby growth measurements can also be stored over time for monitoring development. MARGAM additionally supports one-to-one communication between mothers and their assigned ASHA Workers, providing a direct channel for follow-up and assistance. Overall, the system integrates maternal records, baby registration, vaccination management, growth tracking, appointments, notifications, health information and communication within a single web application, providing a structured approach to healthcare coordination.

2.3 Feasibility Study

A feasibility study evaluates whether the proposed MARGAM system can be successfully developed, implemented and maintained using the available technologies, resources and development environment. It assesses the technical, economic and operational suitability of the system while considering its intended users, requirements and limitations. For MARGAM, the feasibility analysis focuses on technical feasibility, economic feasibility, operational feasibility and the planned development schedule to determine whether the system can achieve its objectives within the academic development environment.

2.3.1 Technical Feasibility

Technical feasibility examines whether the required technologies and resources are available and suitable for developing the system. MARGAM is developed as a browser-based web application using Python and Django, with SQLite for database management and HTML, CSS, JavaScript and Bootstrap 5 for the user interface. These technologies support authentication, database operations, role-based access, dashboards, forms and healthcare workflows. Django provides a structured architecture for managing users, mothers, babies, vaccinations, growth measurements, appointments and notifications, while security mechanisms such as CSRF protection, object-level authorization and controlled state-changing operations support secure implementation. Since these technologies are widely available, lightweight and suitable for academic development, the technical implementation of MARGAM is feasible and practical.

2.3.2 Economic Feasibility

Economic feasibility determines whether the proposed system can be developed and maintained within the available financial resources. MARGAM primarily uses open-source and freely available technologies such as Python, Django, SQLite and Bootstrap 5, eliminating the need for expensive software licences. The system does not require specialized hardware or costly infrastructure for its academic development, testing or demonstration and can be implemented using standard computer systems. SQLite further reduces infrastructure requirements during development. Therefore, MARGAM is economically feasible for academic implementation, while its current architecture also provides a practical foundation for future expansion with additional infrastructure when required.

2.3.3 Operational Feasibility

Operational feasibility evaluates whether the system is practical, useful and easy for its intended users to operate. MARGAM provides separate role-based dashboards for Government Officials, ASHA Workers, Doctors and Mothers, with each user receiving access to functions relevant to their responsibilities. The system supports structured workflows for mother and baby management, vaccination tracking, growth monitoring and appointment handling. Automated vaccination record generation reduces repetitive work, while appointment validation, controlled status changes, notifications and role-based access improve consistency and usability. These features enable different stakeholders to manage their activities through a coordinated platform while accessing only the information and functions appropriate to their roles.

CHAPTER 3

LITERATURE REVIEW

A literature survey was carried out to understand how earlier researchers and developers have approached the key areas addressed by MARGAM, including digital maternal and child healthcare management, vaccination tracking, healthcare record management, appointment systems, and technology-assisted coordination between healthcare stakeholders. The surveyed works are summarized in the following sections, and their relevance to the present project is discussed. The descriptions are based on the approaches, techniques, and findings reported in the literature reviewed for this project.

3.1 Web-Based Child Immunization Information Management System (2021)

This work, authored by Salawu (2021), focuses on the development of a web-based system for managing child immunization information. The system aims to improve the organization and accessibility of vaccination records by replacing conventional record management with a digital approach.

- **Web-based record management:** Child immunization information is maintained through a centralized web-based system.
- **Vaccination tracking:** The system supports the management and monitoring of vaccination-related information.
- **SMS reminders:** Reminders are provided to help users remember upcoming vaccination activities.
- **Digital accessibility:** Vaccination information can be accessed and managed more conveniently than traditional records.

Relevance: The study demonstrates the usefulness of web-based vaccination management and automated reminders. MARGAM follows a similar approach by maintaining child vaccination records and providing notifications for scheduled vaccination activities.

3.2 VacChain: A Blockchain-Based EMR System to Manage Child Vaccination Records (2022)

This work, authored by Alnssayan et al. (2022), proposes VacChain, a blockchain-based Electronic Medical Record system for managing child vaccination records. The system focuses on improving the security, integrity and reliability of digital vaccination information.

- **Electronic medical records:** Child vaccination information is stored digitally.
- **Blockchain technology:** Blockchain is used to improve the security and integrity of stored records.
- **Secure record management:** The system aims to protect vaccination information from unauthorized modification.
- **Reliable information:** Digital records provide a structured and reliable vaccination history.

Relevance: The study highlights the importance of secure digital healthcare records. MARGAM also provides role-based access to maternal and child healthcare information so that only authorized users can manage relevant records.

3.3 Vaccine Management System (2023)

This work, developed by Kamal Acharya (2023), presents a digital Vaccine Management System for organizing vaccination-related activities. The system focuses on vaccination scheduling, management and reporting to improve the monitoring of vaccination services.

- **Vaccine management:** Vaccination information is maintained through a digital system.
- **Schedule management:** Vaccination schedules can be organized and monitored.
- **Reporting:** Reports can be generated from vaccination-related information.
- **Monitoring:** The system supports better monitoring of vaccination activities.

Relevance: The study demonstrates the benefits of digital vaccination scheduling and reporting. MARGAM extends this approach by generating vaccination records automatically based on predefined vaccination schedules when a baby is registered.

3.4 Digitalization of Home-Based Records for Maternal, Newborn, and Child Health (2023)

This work, authored by Geldof M., Gerlach N. and Portela A. (2023), focuses on the digitalization of home-based records related to maternal, newborn and child health. The study highlights the importance of converting traditional healthcare records into digital formats.

- **Digital health records:** Maternal, newborn and child health information is maintained electronically.
- **Improved accessibility:** Digital records can be accessed more easily when required.
- **Organized information:** Healthcare information can be stored in a structured format.
- **Continuity of records:** Digital records support the maintenance of healthcare information over time.

Relevance: The study is closely related to MARGAM because the system maintains digital maternal and child records. MARGAM further connects these records with vaccination, appointments, growth information and notifications.

3.5 Digital Health Technologies for Maternal and Child Health (2023)

This work, authored by Till S. et al. (2023), examines the application of digital health technologies in maternal and child healthcare. The study highlights how digital technologies can improve access to healthcare information and support healthcare services.

- **Digital healthcare:** Digital technologies are applied to maternal and child healthcare.
- **Healthcare information:** Digital platforms improve access to relevant health information.

- **Technology-based services:** Digital tools can support healthcare activities and communication.
- **Improved accessibility:** Technology can make healthcare information more accessible to users.

Relevance: The study supports the use of digital technologies for maternal and child healthcare. MARGAM applies this concept through a centralized web platform connecting Mothers, ASHA Workers, Doctors and Government Officials.

3.6 Antenatal Care Management Platform (2023)

This work, authored by Aliyi A., Mesfin B., Hassen F. et al. (2023), presents a digital platform for managing antenatal care activities. The system focuses on organizing pregnancy-related information and supporting maternal healthcare monitoring.

- **Antenatal care management:** Pregnancy-related healthcare activities are managed digitally.
- **Maternal monitoring:** The system supports monitoring of maternal health information.
- **Digital records:** Pregnancy-related information is maintained in an organized manner.
- **Healthcare management:** The platform supports systematic antenatal care activities.

Relevance: The study is relevant to MARGAM because maternal healthcare management is an important part of the system. MARGAM maintains maternal information and connects it with other healthcare services through a centralized platform.

3.7 Vaccidoc: Child Vaccination Management System (2025)

This work, authored by Renuka Kajale, Nazneen Shaikh and Gayatri Nagendre (2025), presents Vaccidoc, a web-based child vaccination management system. The system focuses on managing vaccination information and providing reminders for upcoming vaccination activities.

- **Web-based vaccination management:** Child vaccination information is managed through a web application.

- **Vaccination schedules:** Vaccination schedules are maintained digitally.
- **Reminders:** Users receive reminders about upcoming vaccination activities.
- **Digital records:** Vaccination information can be maintained and accessed electronically.

Relevance: The study demonstrates the usefulness of web-based vaccination management and reminders. MARGAM incorporates similar functionality through vaccination tracking and notifications within the Mother and healthcare-worker portals.

3.8 Digital Health Platform for Maternal Health (2025)

This work, authored by Ajayi T. (2025), focuses on the development of a digital health platform for maternal healthcare. The platform provides facilities for managing maternal health records and appointment-related information.

- **Maternal health records:** Maternal healthcare information is maintained digitally.
- **Appointment management:** Healthcare appointments are managed through the platform.
- **Centralized information:** Relevant maternal healthcare information is maintained in one system.
- **Healthcare coordination:** Digital management supports better organization of maternal healthcare activities.

Relevance: The study demonstrates the benefits of centralized digital maternal healthcare management. MARGAM incorporates maternal records and appointment management while also integrating child healthcare, vaccination, growth monitoring and notifications.

3.9 Digital Maternal Healthcare Management System

This work focuses on the use of digital systems for managing maternal healthcare information and supporting communication between healthcare stakeholders. The approach aims to improve the organization of maternal records and make healthcare information more accessible.

- **Digital maternal records:** Maternal information is maintained electronically.
- **Healthcare coordination:** Digital systems support coordination between users and healthcare providers.
- **Centralized information:** Healthcare information can be maintained in an organized system.
- **Improved accessibility:** Authorized users can access relevant maternal healthcare information.

Relevance: The study supports the concept of centralized maternal healthcare management. MARGAM extends this approach by integrating maternal and child records with vaccination, appointments, notifications and role-based portals.

3.10 Integrated Maternal and Child Healthcare Management System

This work focuses on integrating different maternal and child healthcare activities into a centralized digital platform. The approach demonstrates how digital systems can combine healthcare records and related services to improve coordination between different stakeholders.

- **Centralized healthcare management:** Maternal and child information is managed through a common platform.
- **Role-based access:** Different users can be provided with access according to their responsibilities.
- **Healthcare coordination:** The system supports interaction between healthcare workers and beneficiaries.
- **Integrated services:** Different healthcare activities can be managed within one system.

Relevance: The approach is directly related to MARGAM, which integrates Government Officials, ASHA Workers, Doctors and Mothers through separate role-based portals. The system combines maternal records, baby registration, vaccination management, growth tracking, appointments and notifications in one platform.

3.1 Summary of the Literature Survey

Table 2.1: Summary of the surveyed literature

Work (Year)	Source	Technique / Area	Contribution	Outcome
Web-Based Child Immunization Information Management System (2021)	Salawu	Web system + SMS	Child immunization management and reminders	Improved vaccination tracking
VacChain (2022)	Alnssayan et al.	Blockchain + EMR	Secure vaccination records	Improved record security
Vaccine Management System (2023)	Kamal Acharya	Scheduling + Reporting	Digital vaccine management	Better vaccination monitoring
Digitalization of Home-Based Records (2023)	Geldof et al.	Digital health records	Maternal and child record management	Improved accessibility
Digital Health Technologies for MCH (2023)	Till S. et al.	Digital health technologies	Digital maternal and child healthcare	Improved healthcare access
Antenatal Care Management Platform (2023)	Aliyi et al.	Digital antenatal care	Maternal healthcare monitoring	Better antenatal management
Vaccidoc (2025)	Kajale et al.	Web + Reminders	Child vaccination management	Improved vaccination follow-up
Digital Health Platform for Maternal Health (2025)	Ajayi T.	Digital records + Appointments	Maternal healthcare management	Better healthcare coordination
Digital Maternal Healthcare Management System	Digital healthcare	Digital records	Maternal information management	Improved accessibility
Integrated Maternal and Child Healthcare Management System	Digital healthcare	Centralized + Role-based system	Integrated healthcare management	Better stakeholder coordination

3.12 Research Gap Identified

The survey shows that digital technologies have been widely applied to maternal healthcare, child vaccination, healthcare records and appointment management. However, many existing systems focus on individual healthcare activities rather than providing an integrated platform for all major maternal and child healthcare services.

The following gaps were identified:

- Most systems focus mainly on maternal healthcare or vaccination management rather than integrating both.
- Limited systems provide separate role-based access for Government Officials, ASHA Workers, Doctors and Mothers.
- Maternal records, baby records, vaccination, appointments and growth information are often managed separately.
- Automated generation of vaccination records based on predefined schedules is limited.
- There is a need for a centralized platform that improves coordination between different maternal and child healthcare stakeholders.

MARGAM addresses these gaps by integrating maternal records, baby registration, vaccination management, growth tracking, appointments, notifications and role-based portals for Government Officials, ASHA Workers, Doctors and Mothers within a single web-based platform.

CHAPTER 4

DIAGRAMS

4.1 Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is used to represent the flow of information within a system. It shows how data moves between different processes, data stores and external entities involved in the application. In the MARGAM system, the DFD illustrates how information such as authentication details, maternal records, baby information, vaccination data, appointment requests and communication messages are submitted and processed by the system. It also shows how the processed information is stored and returned to authorized users through the appropriate workflows. The diagram helps in understanding how different components of the MARGAM system interact with each other.

4.1.1 DFD LEVEL 0

The Level 0 Data Flow Diagram, also known as the context diagram, provides a high-level overview of the entire MARGAM system. It represents the system as a single main process and shows how it interacts with the four external entities, namely Government Official, ASHA Worker, Doctor and Mother. The diagram illustrates the major inputs and outputs of the system, including user authentication details, maternal and baby information, vaccination activities, appointment requests, notifications and communication data. It provides a simplified representation of the overall data flow without showing the detailed internal processes of the application.

Additionally, the diagram highlights how the different users provide information and requests to the MARGAM system according to their assigned roles. Government Officials provide administrative and schedule-related information, ASHA Workers manage assigned mothers and babies, Doctors handle availability and appointment-related activities, and Mothers provide personal and baby-related information and request healthcare services. The system processes these inputs and returns appropriate dashboards, records, status information and notifications to the authorized users. Overall, the Level 0 DFD provides a clear representation of how information enters the MARGAM system, how it is processed and how it is delivered to the different external entities.

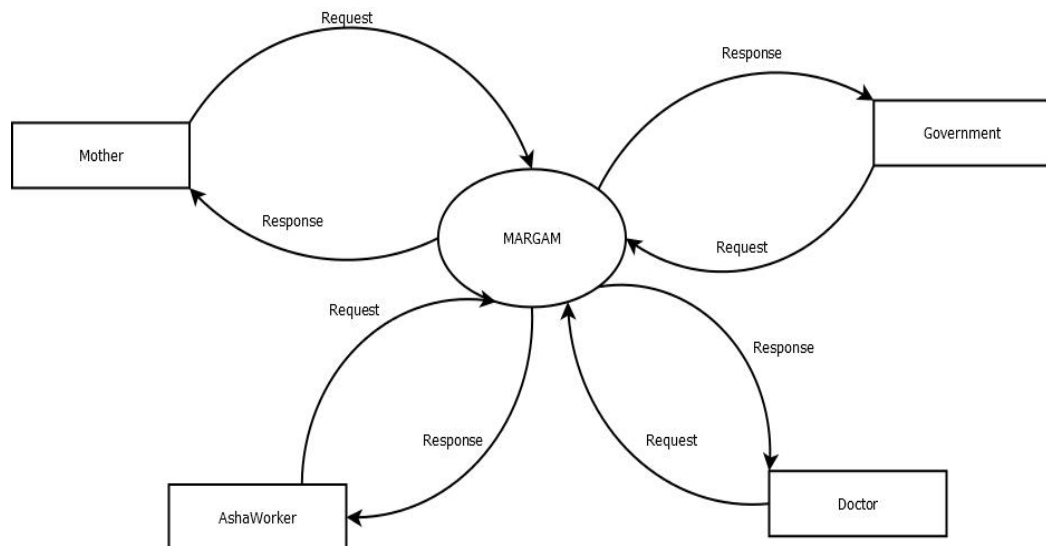


Figure 4.1 DFD Level 0

4.1.2 DFD LEVEL 1: Mother

Figure 4.1.2 shows the main activities performed by the Mother in the MARGAM system, including booking doctor appointments, adding baby profiles, tracking baby growth, viewing vaccination schedules, and communicating with the ASHA Worker.

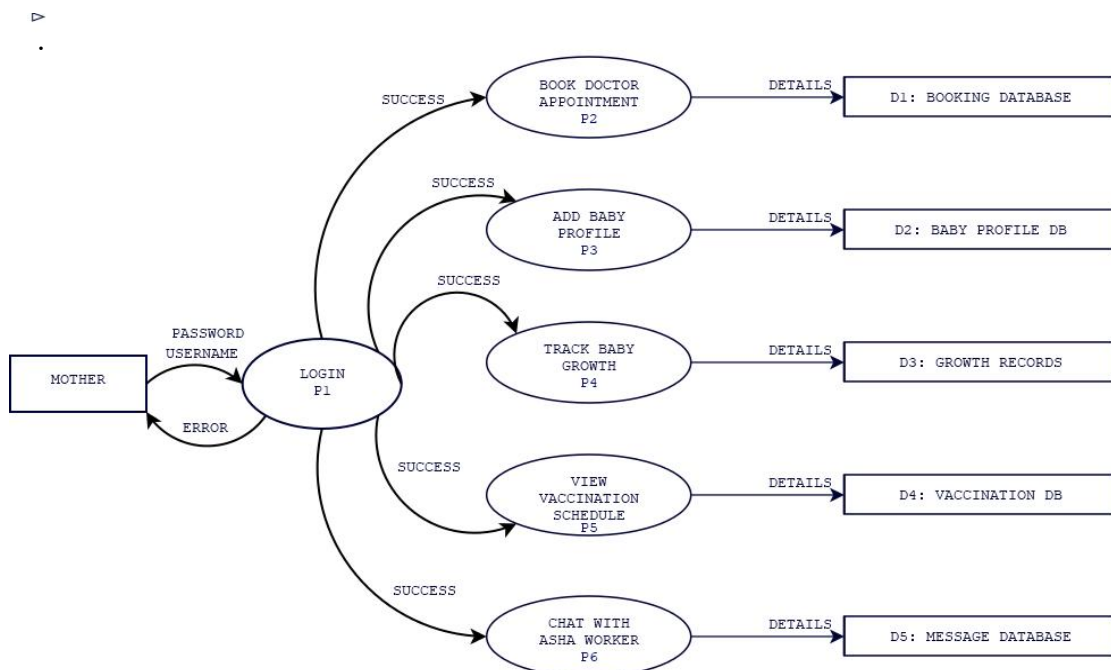


Figure 4.1.2 DFD Level 1: Mother

4.1.3 DFD LEVEL 1: Government

Figure 4.1.3 shows the main activities performed by the Government Official in the MARGAM system, including registering doctors and ASHA Workers, uploading vaccination schedules, managing projects, and sending notifications.

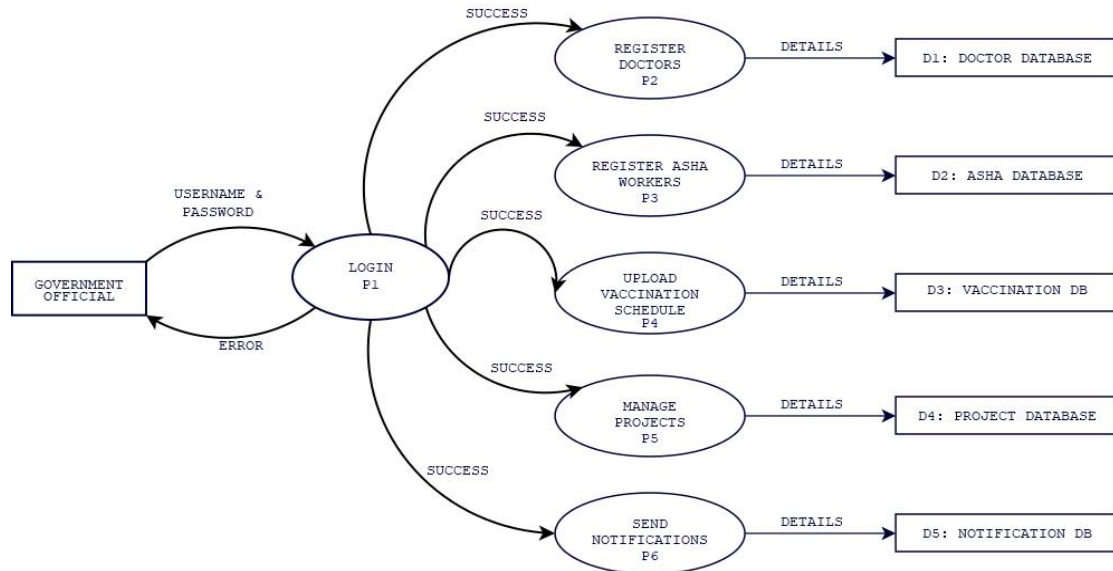


Figure 4.1.3 DFD Level 1: Government

4.1.4 DFD LEVEL 1: Asha Worker

Figure 4.1.4 shows the main activities performed by the ASHA Worker in the MARGAM system, including adding mother and baby information, updating health records, tracking vaccinations, adding health tips and food information, and communicating with mothers.

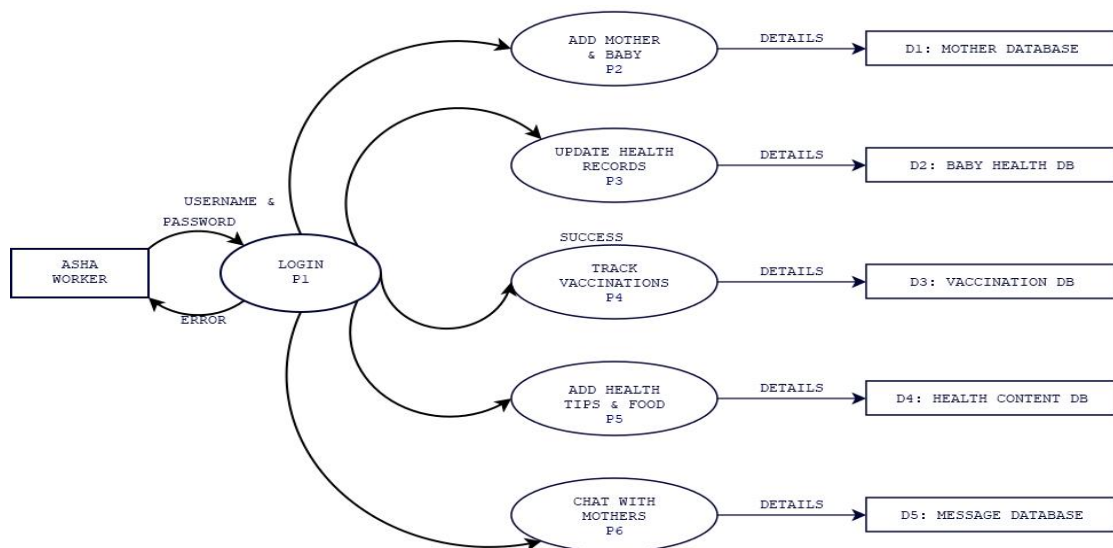


Figure 4.1.4 DFD Level 1: Asha Worker

4.1.5 DFD LEVEL 1: Doctor

Figure 4.1.5 shows the main activities performed by the Doctor in the MARGAM system, including setting availability, managing appointments, viewing patient records, adding post-visit notes, and confirming vaccinations.

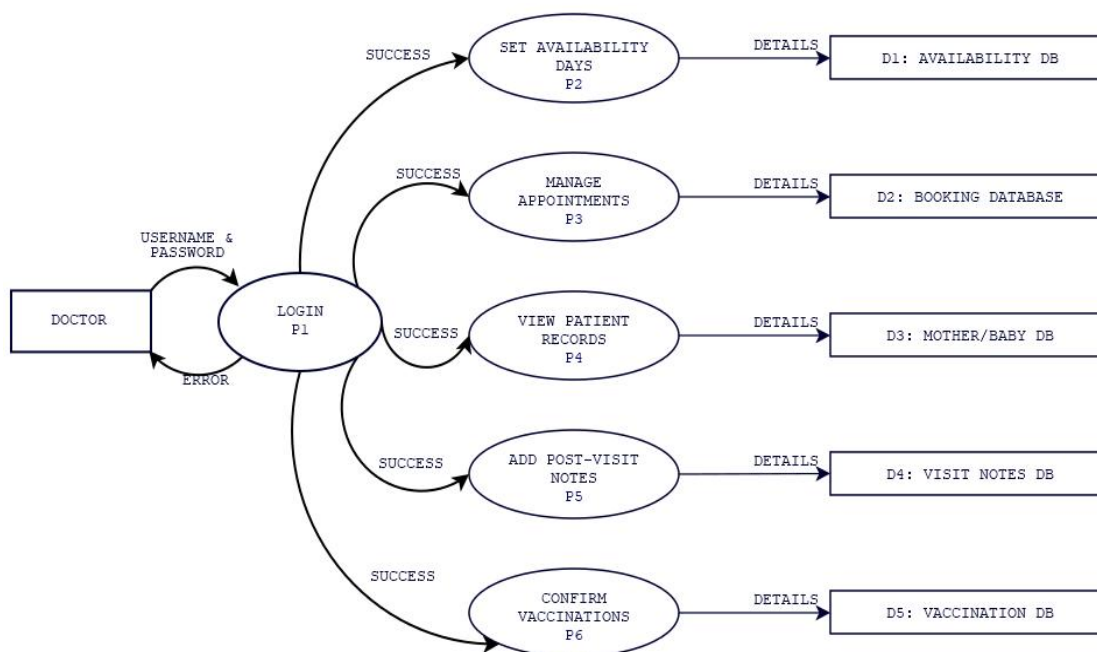


Figure 4.1.5 DFD Level 1: Doctor

4.2 USE CASE DIAGRAM

The use case diagram represents the interaction between the users and the MARGAM system. It shows the main functionalities available in the application and how different actors interact with these features. In this system, the primary actors are Government Official, ASHA Worker, Doctor and Mother. Each actor is associated with a specific set of operations according to the responsibilities and permissions assigned to the corresponding role.

Government Officials can perform administrative activities such as managing doctors and ASHA Workers, managing vaccination schedules, sending notifications and viewing reports. ASHA Workers can manage their assigned mothers and babies, record growth information, follow up vaccination activities and communicate with mothers. Doctors can manage their availability, handle appointment requests and perform authorized appointment actions. Mothers can manage their own information and baby records, request appointments, view vaccination information and communicate with

their assigned ASHA Workers. The diagram helps illustrate the overall functionality of MARGAM and the interaction between each role and the system.

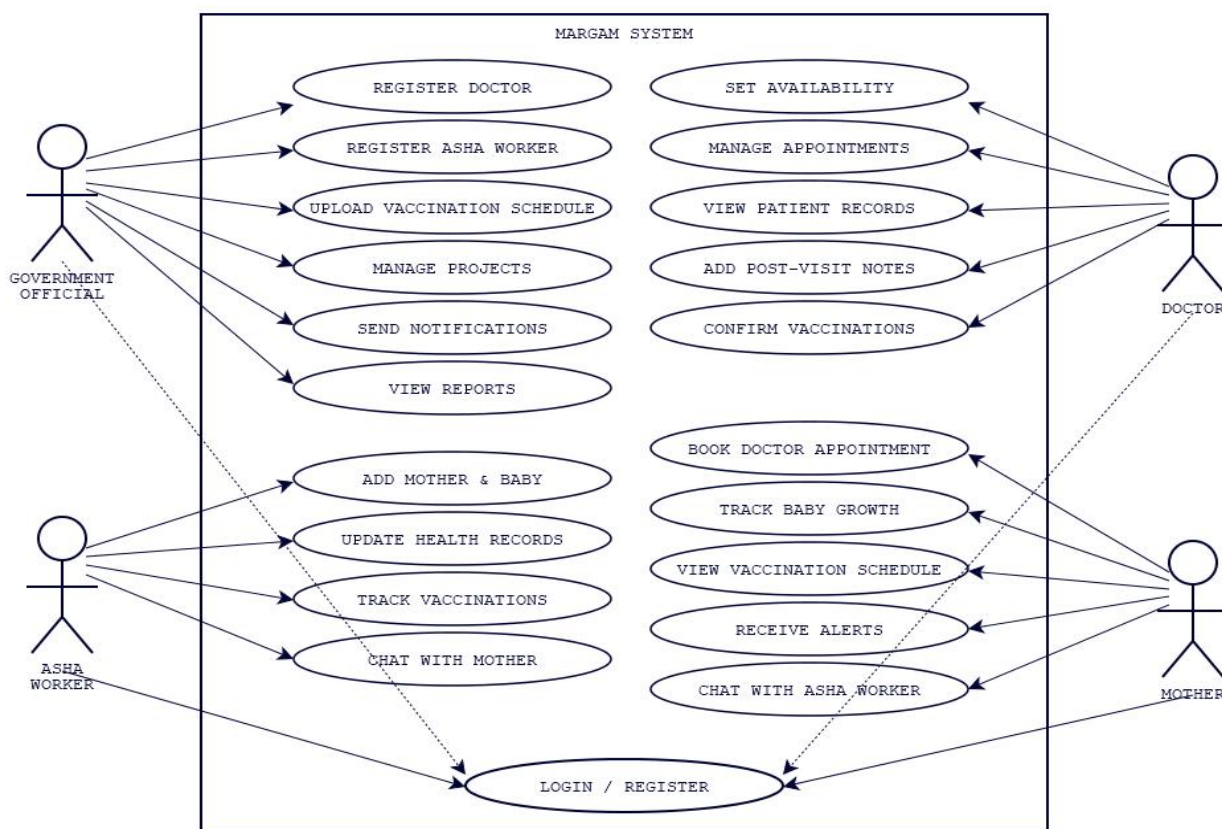


Figure 4.3 Use Case Diagram

4.3 CLASS DIAGRAM

The class or data model represents the structural organization of the MARGAM system by showing the major entities and the relationships between them. The system is organized around a custom user entity and role-specific profiles along with shared healthcare entities. Major entities include CustomUser, Panchayat, MotherProfile, BabyProfile, BabyGrowthRecord, VaccinationScheduleTemplate, VaccinationRecord, Booking, DoctorAvailability, Notification and Message. These entities collectively represent the main information required for managing maternal and child healthcare activities within the application.

The relationships between these entities are important for both functionality and controlled data access. A mother is connected to her registered babies and the ASHA

Worker assigned to her, while a booking connects a mother with a Doctor. Vaccination records are associated with a baby and the corresponding vaccination schedule template, while growth records maintain the historical measurements of a baby. Notifications identify their intended recipients, and messages connect mothers with their assigned ASHA Workers. These relationships allow the application to retrieve and manage information according to actual ownership and professional association rather than relying only on individual identifiers.

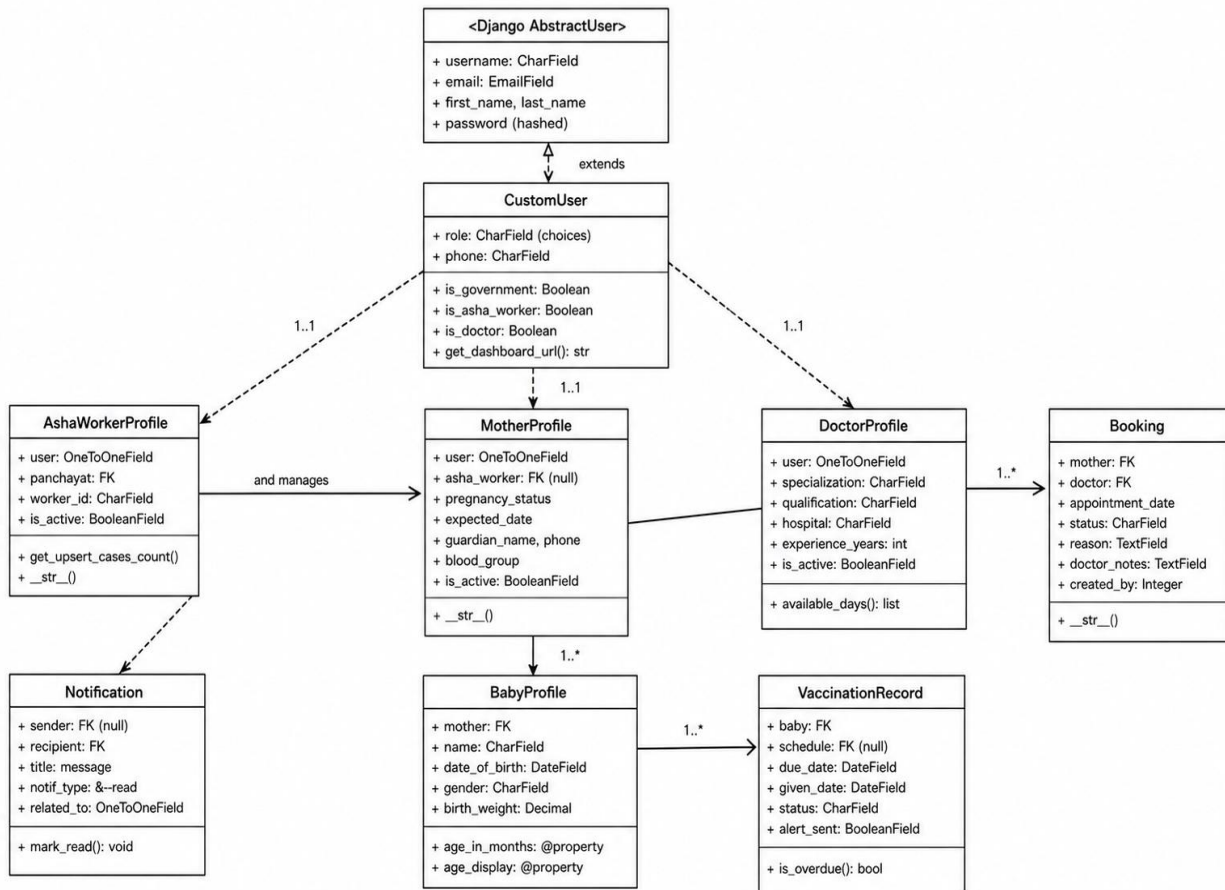


Figure 4.4 Class Diagram

4.4 STATE DIAGRAM

The state diagram represents the different states involved in the appointment lifecycle of the MARGAM system. The process begins when a mother submits an appointment request, placing the booking in the PENDING state. The doctor can then review the request and either confirm or reject it. When the appointment is confirmed, the booking moves to the CONFIRMED state and can later be marked as COMPLETED after the appointment has been completed. A pending appointment may also be cancelled by the mother when the cancellation conditions defined by the system are satisfied.

The state transitions are controlled according to the defined appointment workflow so that users cannot arbitrarily change the status of a booking. Confirmation, rejection and completion are restricted to the appropriate authorized operations, while cancellation is available to the mother for eligible pending appointments. The state diagram therefore provides a clear representation of how an appointment progresses from an initial request through its possible outcomes and ensures that each transition follows the intended workflow of the MARGAM system.

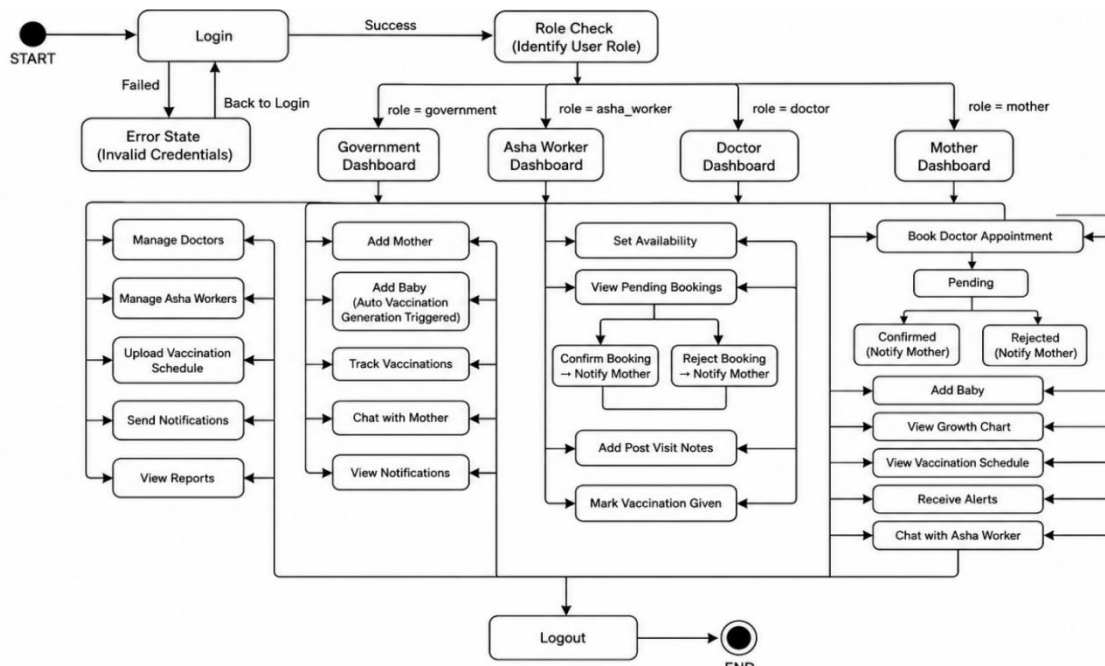


Figure 4.5 State Diagram

4.5 ACTIVITY DIAGRAM

The activity diagram illustrates the workflow of vaccination management in the MARGAM system by showing the sequence of activities involved in creating and maintaining vaccination records. The process begins when an authorized Government Official defines and manages vaccination schedule templates. When a baby is registered in the system, the active vaccination schedule is used to generate the applicable vaccination records for that baby. These records are then maintained as part of the baby's vaccination information and can be accessed by users according to their assigned permissions.

After the vaccination records are generated, authorized users can view the applicable vaccination information and update eligible records as the baby's vaccinations are completed. The activity flow demonstrates the connection between the administrative configuration of vaccination schedules and the operational vaccination records of individual babies. This automated workflow reduces repetitive data entry and ensures

that vaccination information remains organized and connected to the corresponding baby throughout the healthcare process.

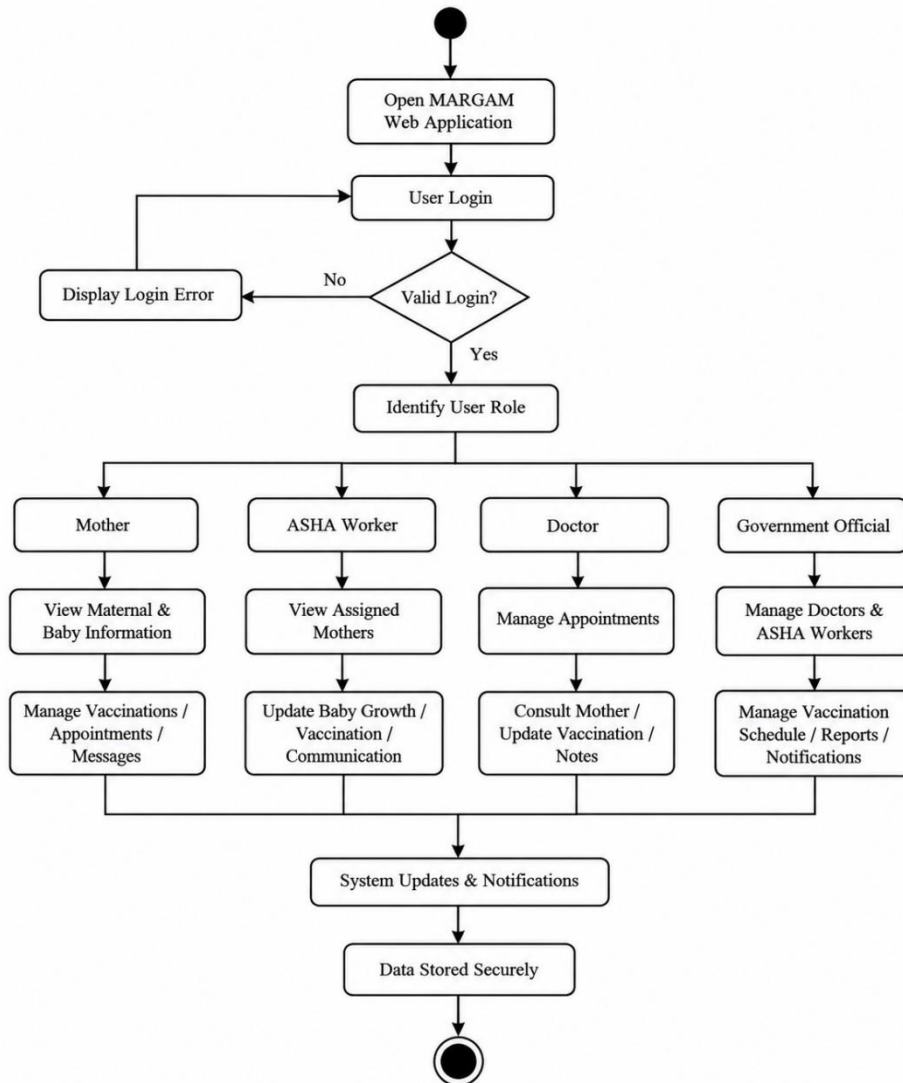


Figure 4.6 Activity Diagram

4.6 SEQUENCE DIAGRAM

The sequence diagram illustrates the interaction between the Mother, the MARGAM application, the doctor's availability information and the booking process during appointment scheduling. The sequence begins when the Mother selects an appointment and submits a request through the application. The system first validates the selected date against the doctor's configured availability and then checks the permitted appointment capacity. Only after these validations are successfully completed does the system create the appointment booking in the appropriate state.

Once a valid pending booking is created, the Doctor can access the request and perform the next authorized action, such as confirming or rejecting the appointment.

The resulting status is then reflected in the system and relevant notification information can be provided to the intended recipient. The sequence diagram illustrates the order in which the different components and users interact during appointment booking and shows how validation, booking creation, doctor action and notification are connected within the MARGAM workflow.

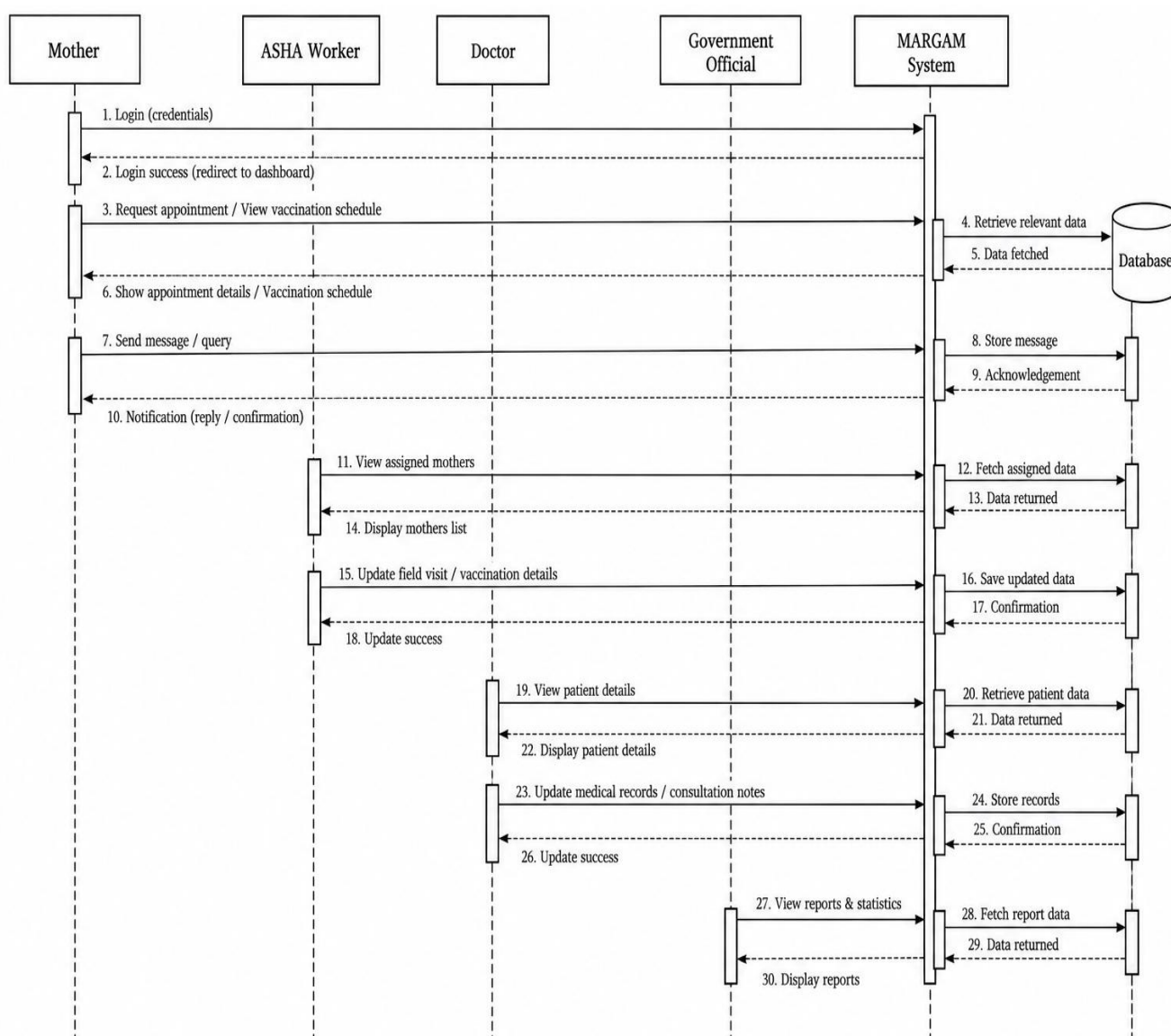


Figure 4.7 Sequence Diagram

CHAPTER 5

METHODOLOGY

5.1 Module Descriptions

MARGAM is organized into role-based modules that provide specific functions for Government Officials, ASHA Workers, Doctors and Mothers. Each user is provided with access according to their responsibilities in the maternal and child healthcare process. The modules work together to maintain healthcare information, manage vaccinations, monitor baby growth, handle appointments and support coordination between the different stakeholders. The major modules of the system are described below.

5.1.1 Authentication Module

The Authentication Module provides the basic entry point to the MARGAM system. It includes the landing page, user login and registration facilities. Mothers can register themselves in the system, while accounts for Government Officials, ASHA Workers and Doctors are managed through authorized access. After successful authentication, users are directed to their respective dashboards according to their assigned roles. This ensures that users can access only the functions relevant to their responsibilities.

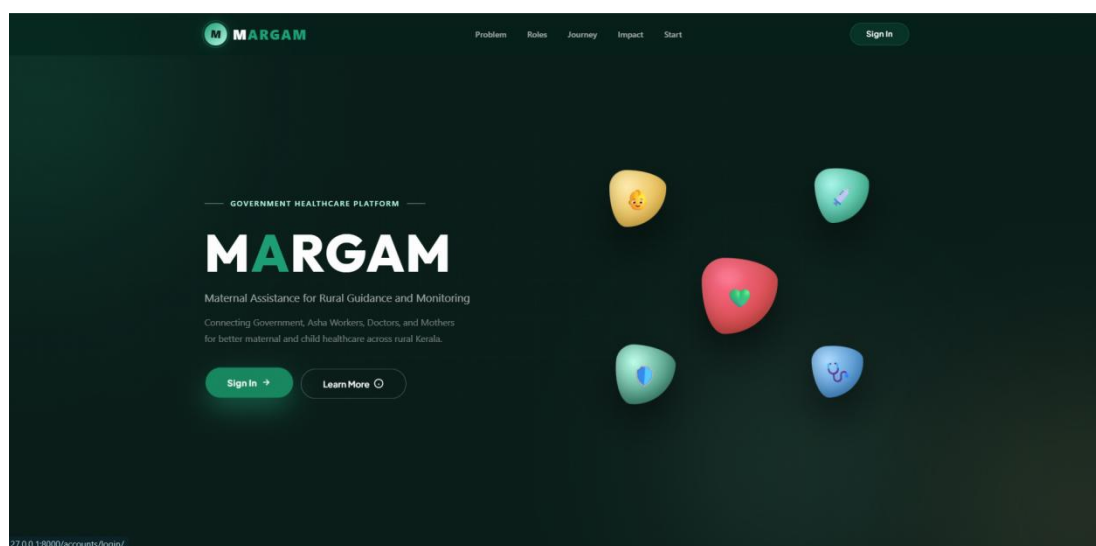


Figure 5.1: MARGAM Landing Page

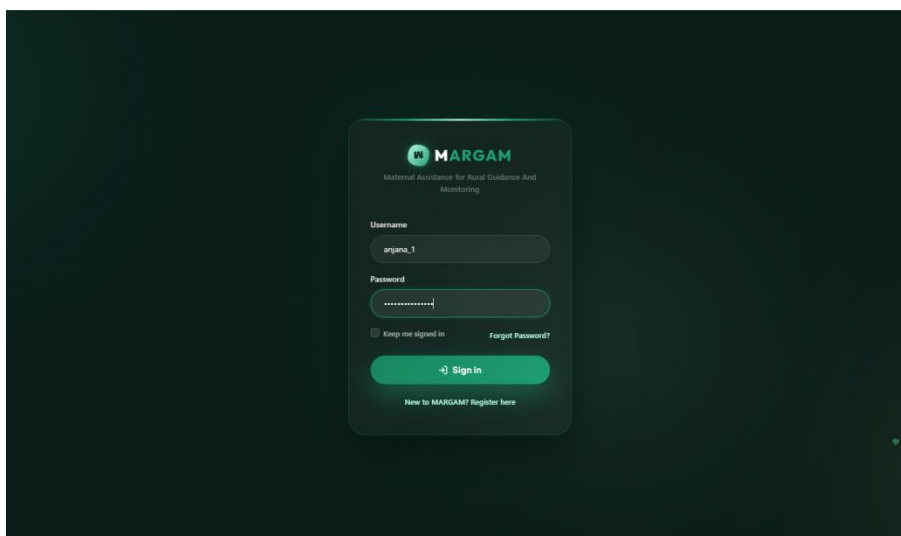


Figure 5.2: Login Interface

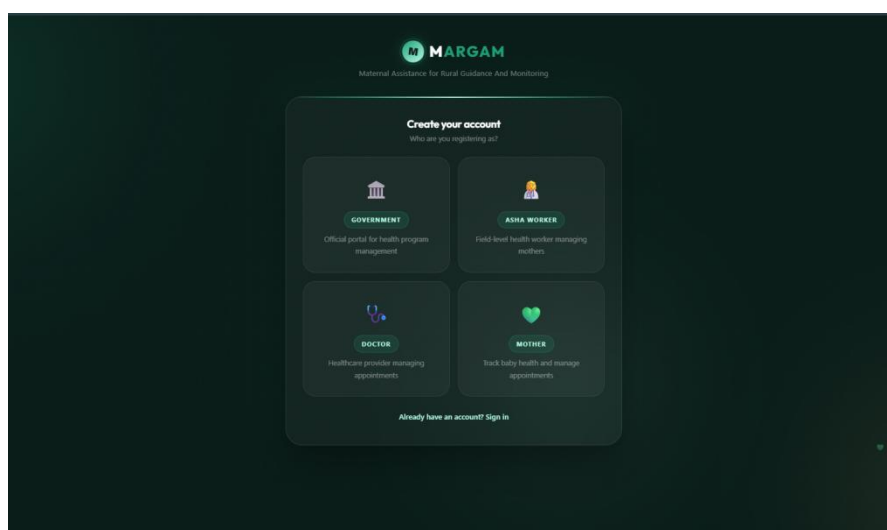


Figure 5.3: Role Selection - Registration

MARGAM
Maternal Assistance for Rural Guidance And Monitoring

MOTHER
Create your account

Username
Username
Required. 150 characters or fewer. Letters, digits and @/+/./ only.

First name
First name

Last name
Last name

Email address
Email address

Phone
Phone

Password
Password

- Your password can't be too similar to your other personal information.
- Your password must contain at least 8 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.

Password confirmation
Password confirmation
Enter the same password as before, for verification.

Date of birth
mm/dd/yyyy

Address
Address

Guardian name
Guardian Name

Guardian phone
Guardian Phone

Pregnancy status
▼

Expected due date
mm/dd/yyyy

Blood group
▼

Create Account

[Back](#) [Already registered? Sign in](#)

Figure 5.4: Mother Self-Registration Form

5.1.2 Government Module

The Government Module provides administrative functions for managing the healthcare resources and activities of the MARGAM system. Government Officials can manage Doctor and ASHA Worker accounts, maintain vaccination schedules, manage healthcare-related information and send notifications. The module also provides reports and dashboard information that help Government Officials monitor the overall activities of the system. These functions support centralized administration and better coordination of maternal and child healthcare services.

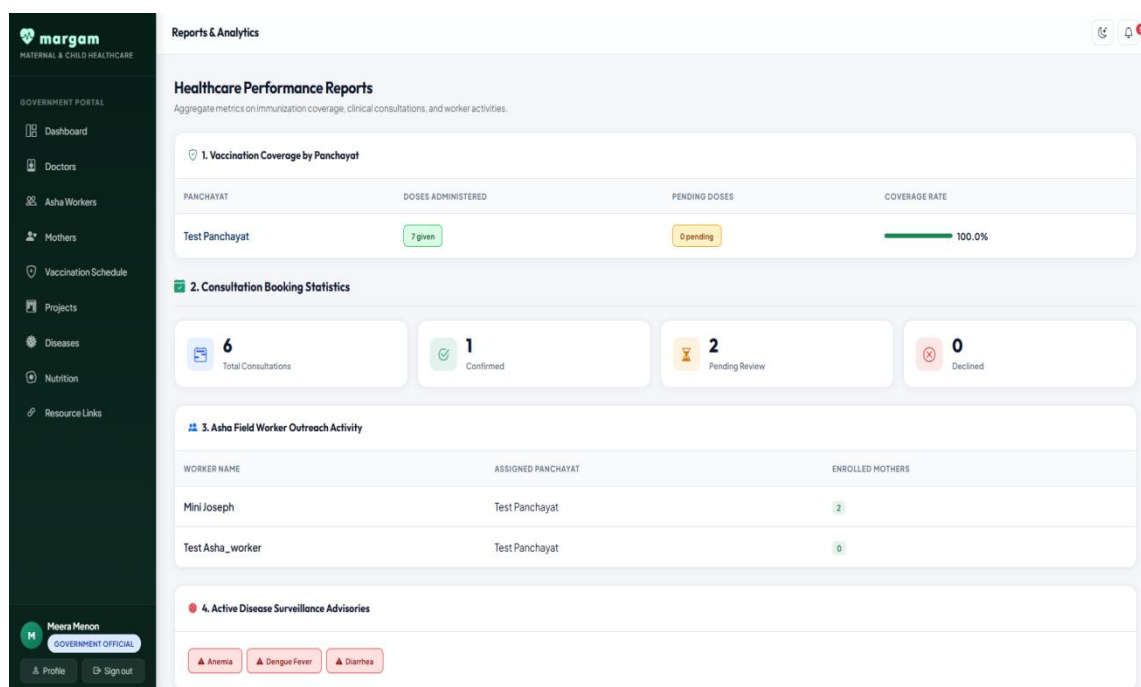


Figure 5.5: Government Dashboard with Key Metrics and Analytics

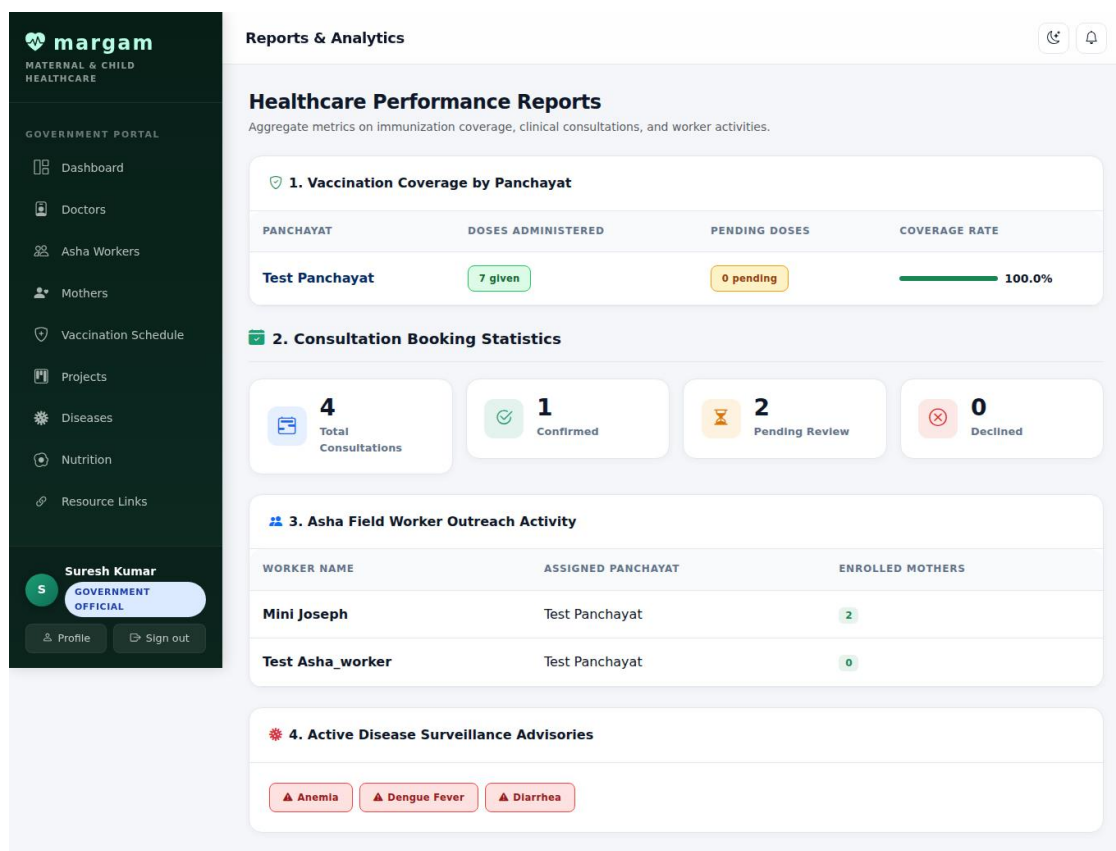


Figure 5.6: Government - Vaccination Report

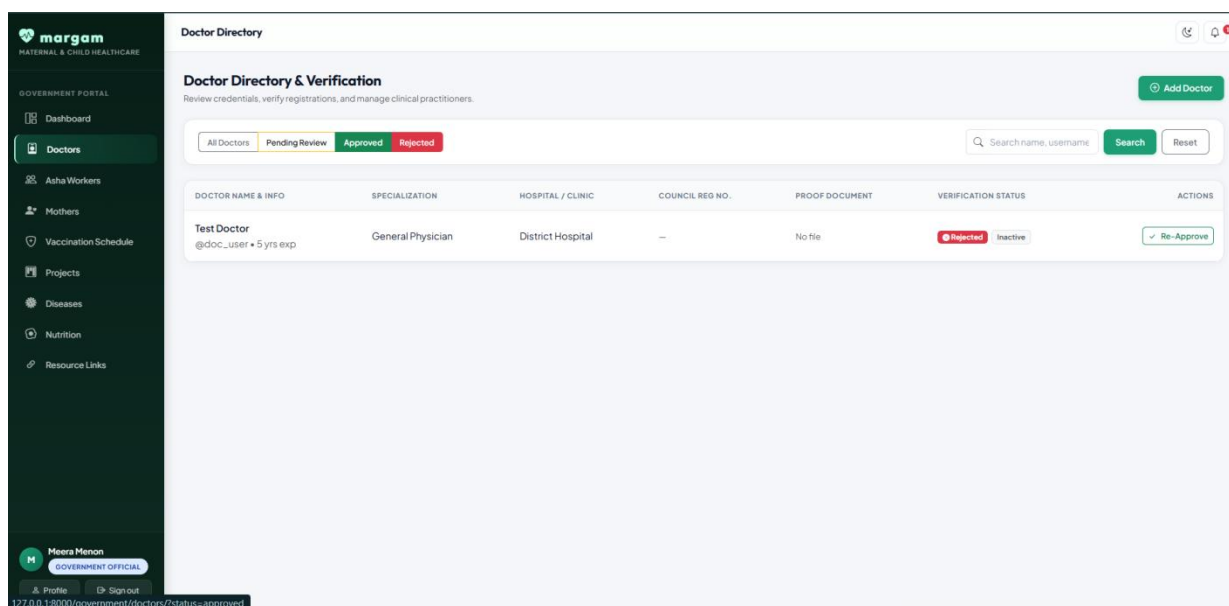


Figure 5.7: Government - Doctor Account Management

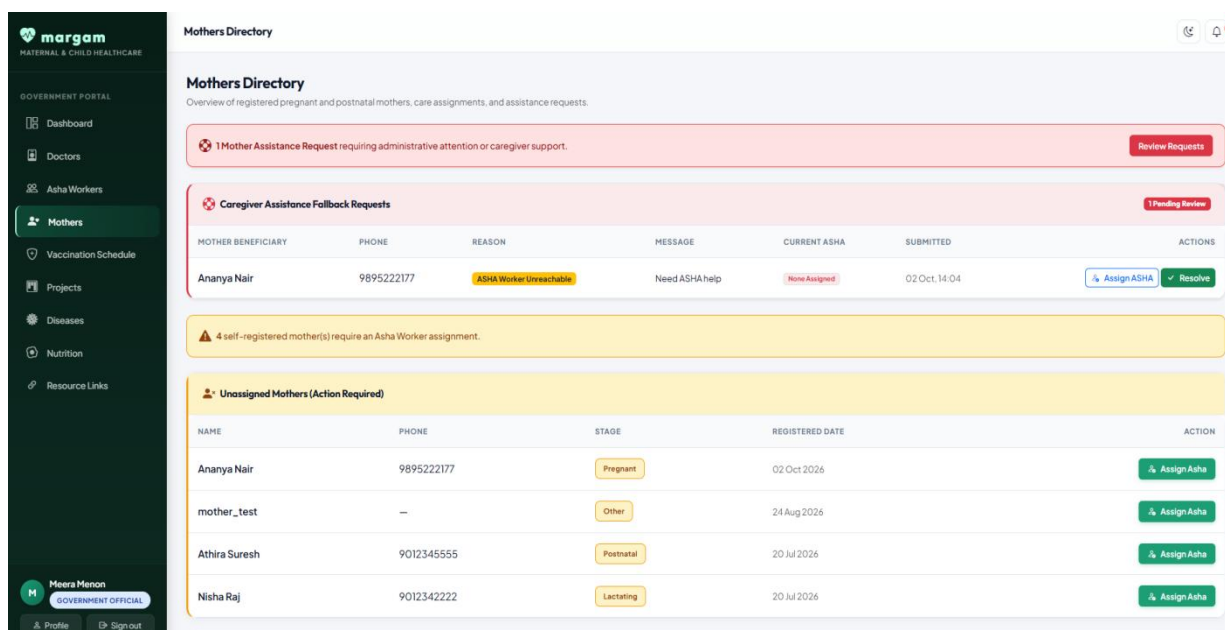


Figure 5.8: Government - Mother, Asha Worker Management

5.1.3 Asha Worker Module

The ASHA Worker Module is designed to support ASHA Workers in managing the mothers and babies assigned to them. The module allows ASHA Workers to view and update maternal and child health information, maintain baby records, track vaccination activities and monitor growth-related information. It also provides facilities for sharing

health tips and food-related information. Communication with mothers allows ASHA Workers to provide follow-up support and maintain better coordination with the beneficiaries.

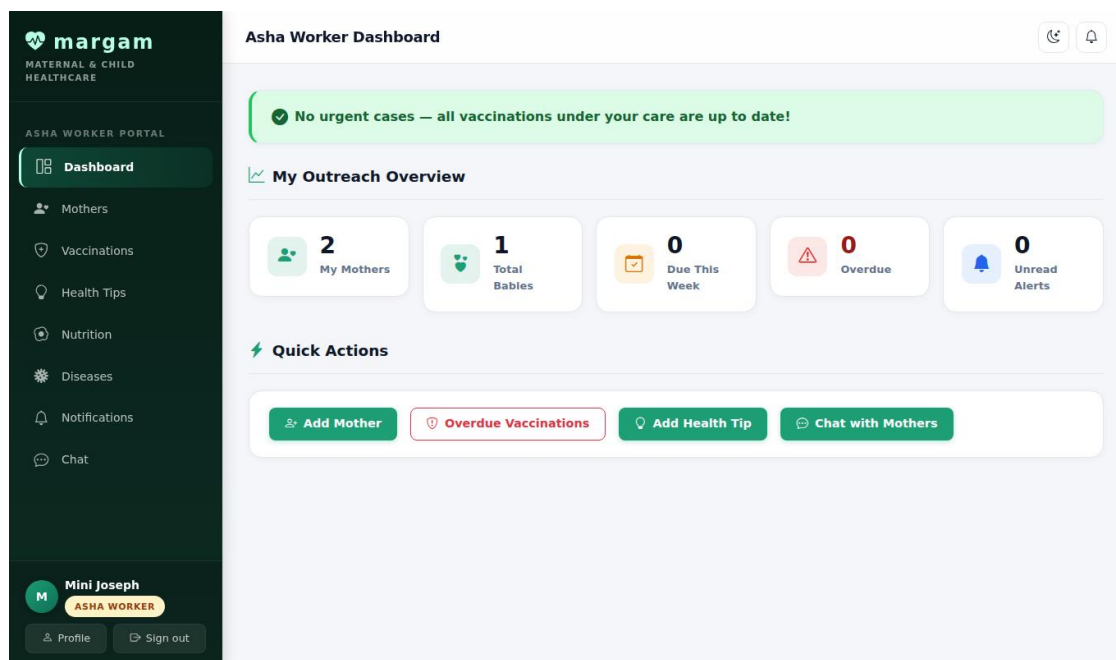


Figure 5.9: Asha Worker Dashboard

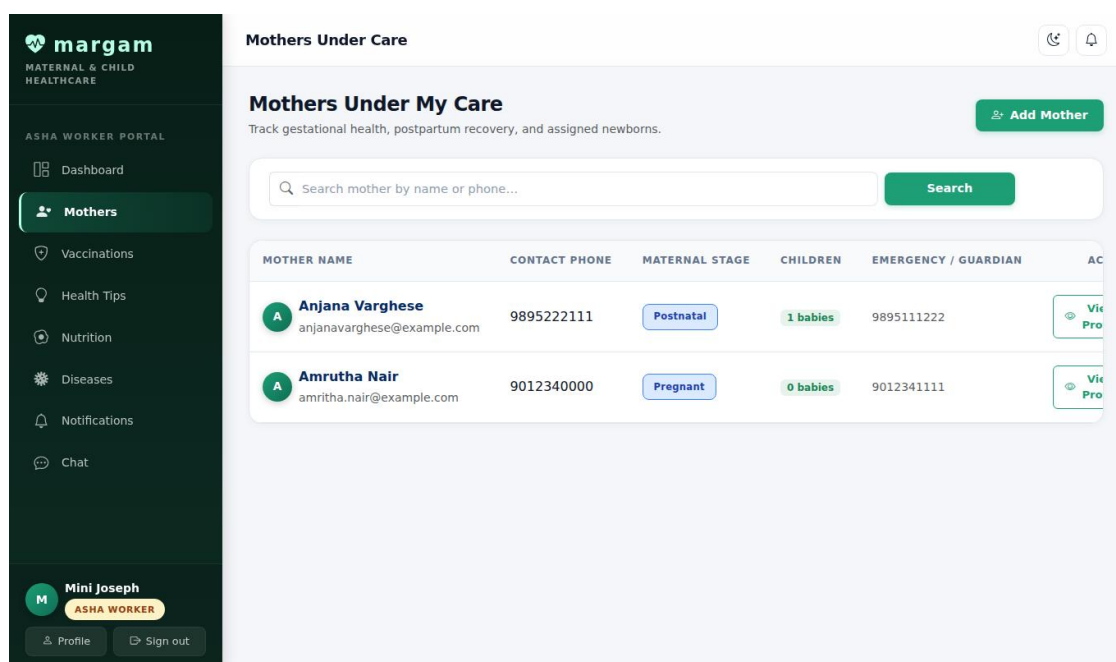


Figure 5.10: Asha Worker - Assigned Mothers List

Mother Profile

← Back to Mothers Directory Add Baby Profile

Anjana Varghese Postnatal B+

PHONE / USERNAME: 9895222111 DATE OF BIRTH: Sep 18, 1998 EXPECTED DUE DATE (EDD): Jul 25, 2026

GUARDIAN / SPOUSE: Nithin Varghese GUARDIAN PHONE: 9895111222 ASSIGNED ASHA WORKER: Mini Joseph

RESIDENTIAL ADDRESS: Green Meadows House, Pampady P.O., Kottayam, Kerala - 686502

Enrolled Children (1) Add Baby

Aarav Varghese Boy

Age: 2 months old

BIRTH DATE: Jul 25, 2026 WEIGHT: 3.20 kg

[View Growth & Vaccines](#)

Figure 5.11: Asha Worker - Mother Profile and Baby Growth Details

Baby Profile

Aarav Varghese Boy

Age: 2 months old

DATE OF BIRTH: Jul 25, 2026 BIRTH WEIGHT: 3.20 kg BLOOD GROUP: B+ MOTHER: Anjana Varghese

[Add Record](#)

Growth Measurements

DATE	WEIGHT (KG)	HEIGHT (CM)	HEAD CIRC. (CM)	RECORDED BY
Jul 25, 2026	3.20 kg	50.0 cm	35.0 cm	Mini Joseph

Immunization Schedule

VACCINE	DOSE	DUE DATE	STATUS	GIVEN DATE	ACTION
BCG	Dose 1	Jul 25, 2026	Green	Jul 25, 2026	Done
OPV-0	Dose 1	Jul 25, 2026	Green	Jun 25, 2026	Done
Hepatitis B	Dose 1	Jul 25, 2026	Green	Jul 25, 2026	Done
Pentavalent	Dose 1	Sep 05, 2026	Green	Sep 05, 2026	Done
OPV	Dose 1	Sep 05, 2026	Green	Sep 05, 2026	Done
IPV	Dose 1	Sep 05, 2026	Green	Sep 05, 2026	Done
Rotavirus	Dose 1	Sep 05, 2026	Green	Sep 05, 2026	Done

Figure 5.12: Asha Worker - Baby Vaccination Management

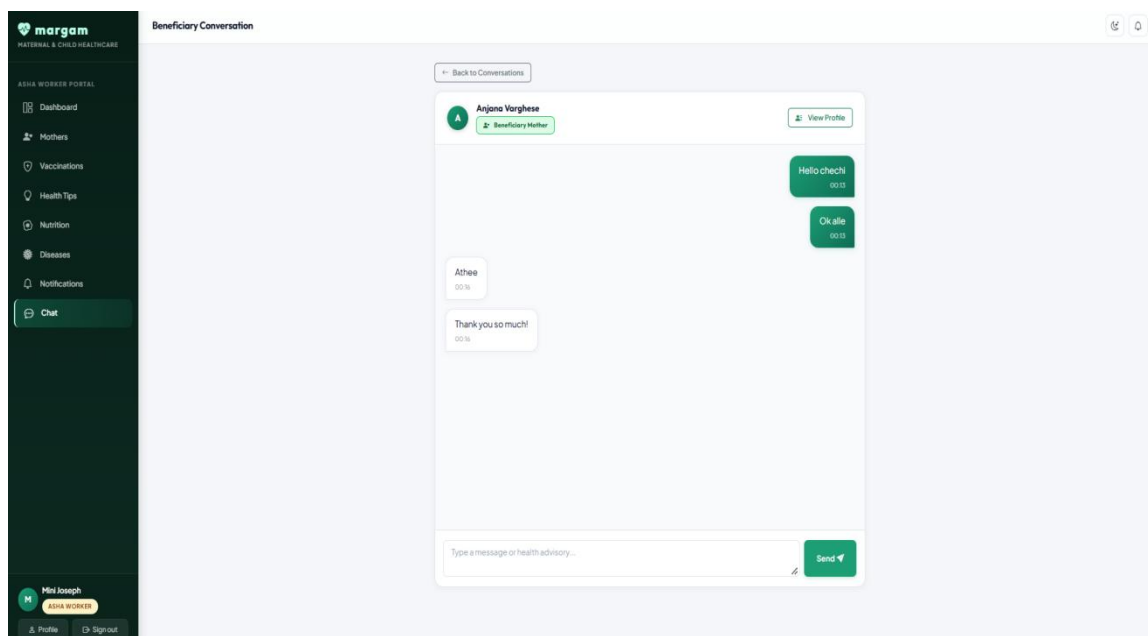


Figure 5.13: Asha Worker - Chat with Mother

5.1.4 Doctor Module

The Doctor Module provides facilities for managing doctor availability and appointments. Doctors can configure their available days, view appointment requests and manage bookings. They can also access relevant patient information, add post-visit notes and update vaccination information when required. This module helps Doctors manage consultations in an organized manner while maintaining the healthcare information associated with the patients.

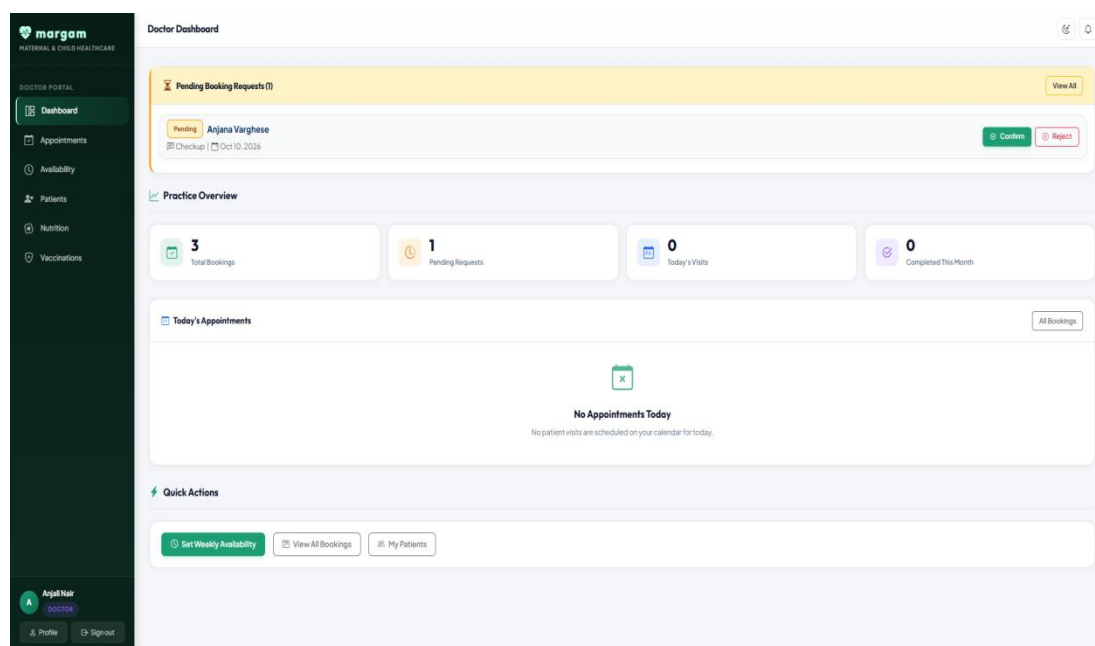


Figure 5.14: Doctor Dashboard with Pending Booking Requests

Set Availability

Weekly Consultation Hours
Set the days and time slots when mothers can book telemedicine or clinic appointments with you.

Weekly Schedule Configuration

Day	Start Time	End Time	Max Patients	Status
Monday	09:00	13:00	8	Available
Tuesday	09:00	17:00	10	Unavailable
Wednesday	09:00	16:00	12	Available
Thursday	09:00	16:00	10	Available
Friday	09:00	17:00	10	Unavailable
Saturday	08:30	16:30	14	Available
Sunday	10:00	14:00	6	Available

Save Availability Schedule

Anjali Nair, DOCTOR
Profile Sign out

Figure 5.15: Doctor - Weekly Availability Configuration

Vaccinations

Patient Immunization Records
Administer and log scheduled childhood vaccines for booked and attending patients.

BABY NAME	MOTHER	VACCINE & DOSE	DUE DATE	STATUS	ACTION
Aarav Varghese	Anjana Varghese	BCG (Dose 1)	Jul 25, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	OPV - O (Dose 1)	Jul 25, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	Hepatitis B (Dose 1)	Jul 25, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	Pentavalent (Dose 1)	Sep 05, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	OPV (Dose 1)	Sep 05, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	IPV (Dose 1)	Sep 05, 2026	Given	Administered
Aarav Varghese	Anjana Varghese	Rotavirus (Dose 1)	Sep 05, 2026	Given	Administered

Anjali Nair, DOCTOR
Profile Sign out

Figure 5.16: Doctor - Vaccination Records of Patients

5.1.5 Mother Module

The Mother Module provides Mothers with access to their maternal and child healthcare information. Mothers can manage their baby information, view vaccination schedules and growth records, and monitor the healthcare activities associated with their babies. The module also allows Mothers to book appointments with Doctors and receive relevant notifications. Communication with the assigned ASHA Worker provides an additional facility for follow-up and healthcare-related assistance.

Main functions: Manage profile and babies, record and view growth information, view vaccination schedules and records, request and cancel eligible appointments, access health information, receive notifications and communicate with the assigned ASHA Worker

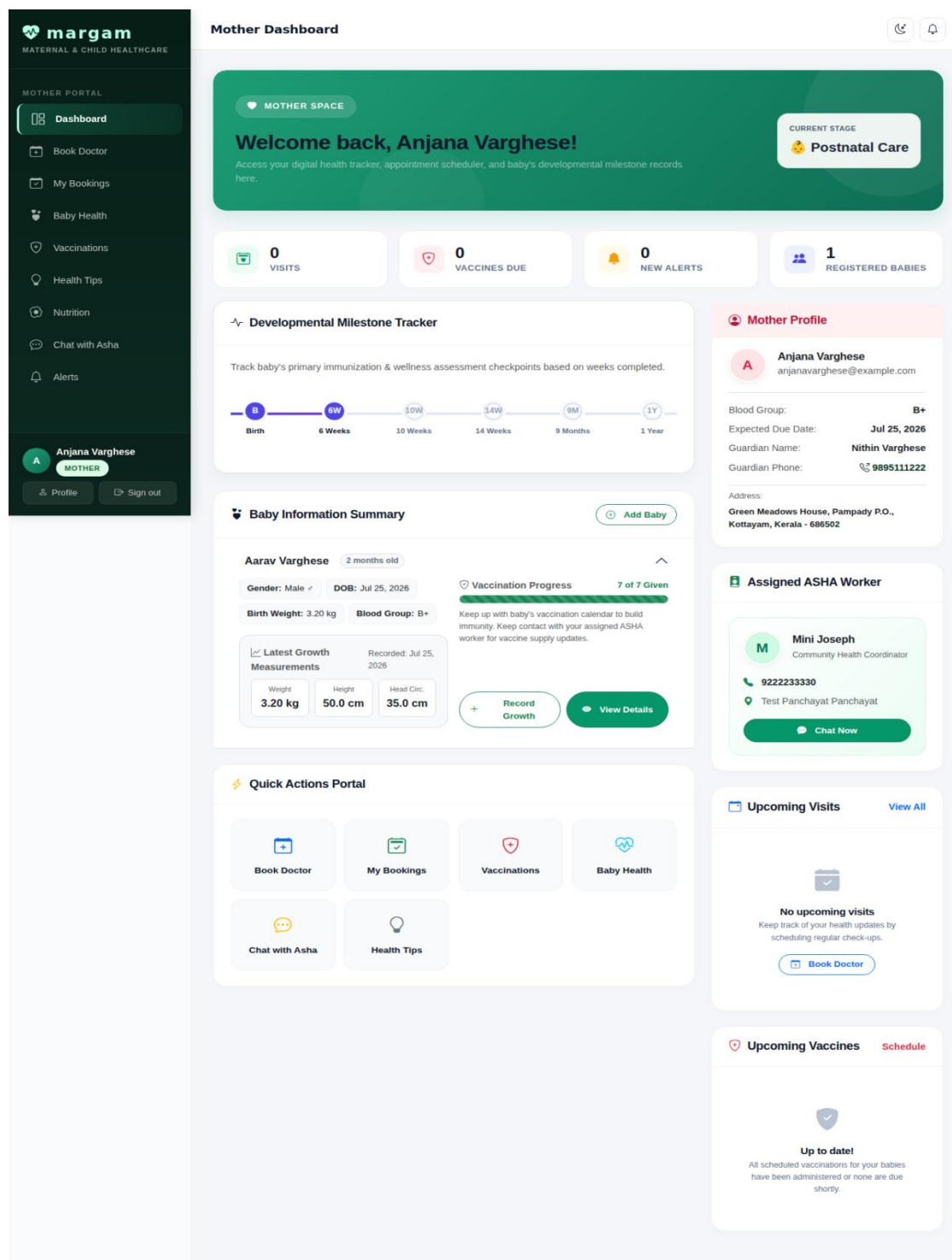


Figure 5.17: Mother Dashboard - Postnatal View

MARGAM
MATERNAL & CHILD
HEALTHCARE

MOTHER PORTAL
Dashboard
Book Doctor
My Bookings
Baby Health
Vaccinations
Health Tips
Nutrition
Chat with Asha
Alerts

Anjana Varghese
MOTHER
Profile Sign out

Book Consultation

1. Select a Doctor

6 available

Dr. Test Doctor
General Physician
District Hospital
Select Doctor

Dr. Anjali Nair
General Physician
St. Mary's Medical Centre
Selected

Dr. Neethu Thomas
Pediatrics
Sunrise Children's Hospital
Select Doctor

Dr. Lakshmi Pillai
Gynecology & Obstetrics
Care Women's Hospital
Select Doctor

Dr. Fathima Ali
Nutrition
Wellness Nutrition Centre
Select Doctor

Dr. doc.test
General Physician
Hospital not listed
Select Doctor

2. Schedule Appointment with Dr. Anjali Nair

SELECTED PRACTITIONER
Dr. Anjali Nair — General Physician
St. Mary's Medical Centre

Preferred Appointment Date
mm/dd/yyyy
Doctor operates on: Monday, Wednesday, Thursday, Saturday, Sunday
Choose an available appointment date for your visit.

Reason for Consultation
Describe symptoms, pregnancy checkup requirements, or questions...

Change Doctor Confirm Booking Request

Figure 5.18: Mother - Book Doctor Appointment

MARGAM
MATERNAL & CHILD
HEALTHCARE

MOTHER PORTAL
Dashboard
Book Doctor
My Bookings
Baby Health
Vaccinations
Health Tips
Nutrition
Chat with Asha
Alerts

Anjana Varghese
MOTHER
Profile Sign out

Baby Detail

Back to Baby Profiles

Add Growth Record

Aarav Varghese Boy
Age: 2 months old

DATE OF BIRTH
Jul 25, 2026

CURRENT AGE
2 months old

GENDER
Male

BIRTH WEIGHT
3.20 kg

Growth Trends (Weight & Height)

Log Measurement

DATE	WEIGHT (KG)	HEIGHT (CM)	NOTES / REMARKS
Jul 25, 2026	3.20 kg	50.0 cm	Healthy full-term newborn. Birth measurements recorded. Baby is active, breastfeeding well, and shows no signs of complications. Initial assessment is normal.

Immunization Milestones

VACCINE NAME	SCHEDULED DUE DATE	STATUS	ADMINISTERED DATE
BCG	Jul 25, 2026	Given	✓ Jul 25, 2026
OPV-0	Jul 25, 2026	Given	✓ Jun 25, 2026
Hepatitis B	Jul 25, 2026	Given	✓ Jul 25, 2026
Pentavalent	Sep 05, 2026	Given	✓ Sep 05, 2026
OPV	Sep 05, 2026	Given	✓ Sep 05, 2026
IPV	Sep 05, 2026	Given	✓ Sep 05, 2026
Rotavirus	Sep 05, 2026	Given	✓ Sep 05, 2026

Figure 5.19: Mother - Baby Health Detail and Vaccination Progress

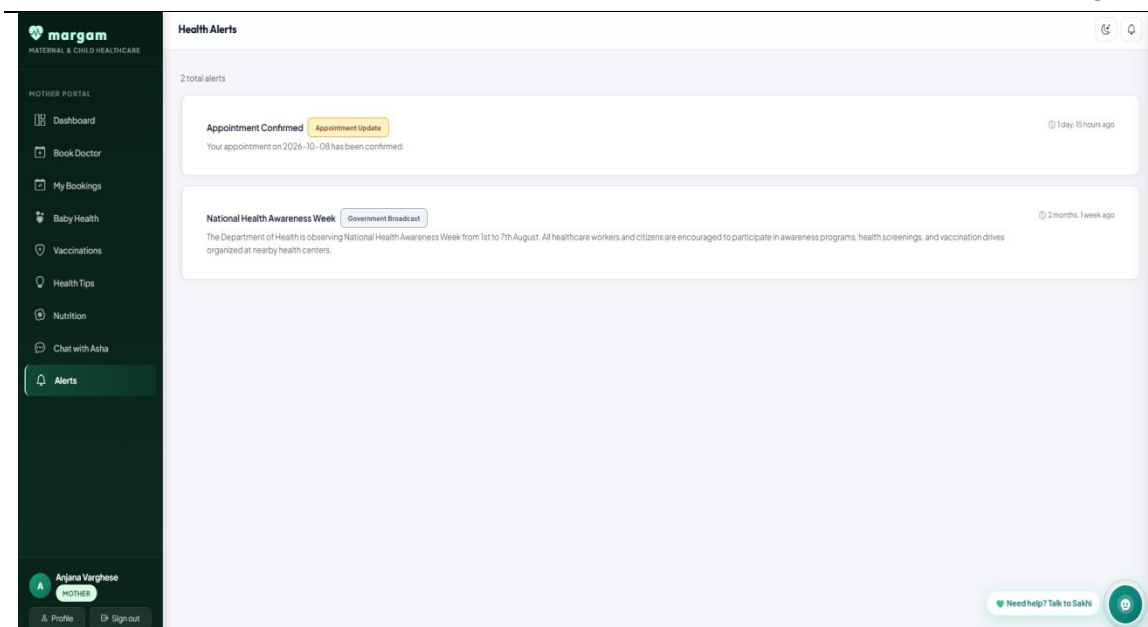


Figure 5.20: Mother - Health Alerts

5.2 Algorithms

MARGAM uses several algorithms and logical procedures to support the major activities of the healthcare platform. These algorithms help manage user authentication, baby registration, vaccination records and appointment booking. The following algorithms describe the basic processing approach used by the system.

User Authentication Algorithm

The authentication process verifies the login details provided by the user and identifies the user's role in the MARGAM system.

Steps:

1. Accept the user's login credentials.
2. Validate the entered information.
3. Check the credentials against the stored user details.
4. If the details are valid, authenticate the user.
5. Identify the user's role.
6. Redirect the user to the corresponding dashboard.
7. If the credentials are invalid, display an appropriate error message.

Automatic Vaccination Record Generation Algorithm

The vaccination record generation process automatically creates vaccination records for a newly registered baby based on the active vaccination schedule.

Steps:

1. Accept the baby's registration details.
2. Store the baby's date of birth.
3. Retrieve the active vaccination schedule.
4. Identify the vaccinations applicable to the baby.
5. Calculate the expected due dates.
6. Create and store the vaccination records.
7. Make the vaccination schedule available to the authorized users.

Appointment Booking Algorithm

The appointment booking process allows a Mother to request an appointment with an available Doctor.

Steps:

1. Select a Doctor and preferred appointment date.
2. Check the Doctor's availability.
3. Check whether the selected slot is available.
4. If the slot is available, create the appointment request.
5. Store the appointment details.
6. Notify the Doctor about the appointment request.
7. If the slot is unavailable, display an appropriate message.

Role-Based Access Control Algorithm

The role-based access process ensures that users can access only the functions permitted for their assigned roles

Steps:

1. Authenticate the user.

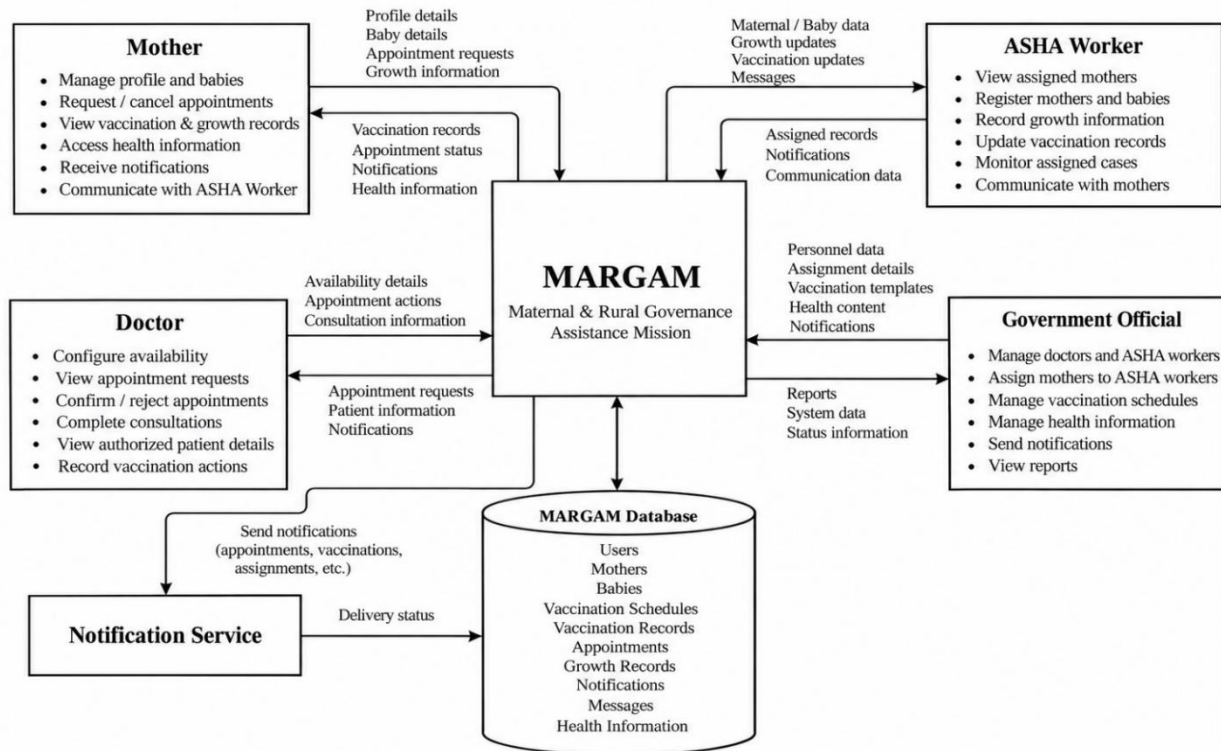
2. Identify the user's assigned role.
3. Check the permissions associated with the role.
4. Provide access to the corresponding dashboard.
5. Allow access to the permitted functions.
6. Restrict unauthorized functions.
7. Display an appropriate access message when permission is denied.

5.3 Architecture

The architecture of the MARGAM system is designed using a web-based client–server architecture in which users access the application through a web browser while the Django framework provides the backend infrastructure. The architecture enables the system to manage maternal and child healthcare information, user authentication, role-based access and healthcare workflows through a centralized platform. The overall design focuses on providing a structured user interface, secure data management and reliable coordination between the different users involved in maternal and child healthcare.

In this architecture, the web browser serves as the primary interface through which users interact with the MARGAM system. The browser provides the user interface for the four main modules of the application: Mother, ASHA Worker, Doctor and Government Official. Users can perform the functions available to their assigned role through the corresponding dashboard. The presentation layer uses Django templates along with HTML, CSS, Bootstrap and JavaScript to provide the different pages, forms, dashboards and interactive elements required by the application.

The Django framework functions as the main backend of the MARGAM system. It handles user authentication, request processing, business logic, role-based authorization and communication with the database. When a user logs into the system, Django verifies the authentication information and determines the user's assigned role. Based on this role, the user is directed to the appropriate module and is provided access only to the functions permitted for that role. This provides a common backend while maintaining separate responsibilities for the four modules.



The backend also manages the healthcare workflows and relationships between different records. Information related to mothers, babies, ASHA Workers, Doctors, vaccinations, appointments, notifications and other healthcare activities is processed through the Django application. The relationship between these records allows the system to maintain connected information instead of treating each record as an isolated entity. For example, a mother can be associated with her registered babies and assigned ASHA Worker, while appointments connect mothers with Doctors and vaccination records are associated with individual babies.

The database layer is implemented using SQLite and is accessed through the Django Object-Relational Mapping (ORM) system. The database stores the information required for the operation of the application, including user details, mother profiles, baby profiles, growth records, vaccination schedules, vaccination records, doctor availability, appointments and notifications. The ORM provides a structured way for the application to create, retrieve, update and manage these records while maintaining the relationships between different entities.

One of the important components of the architecture is the role-based access mechanism. The system identifies whether the authenticated user is a Mother, ASHA Worker, Doctor

or Government Official and provides access to the corresponding functions. In addition to role-based restrictions, access to specific records is controlled according to the user's relationship with the information. For example, an ASHA Worker can access assigned mothers, while a Mother can access her own information and related baby records. Similarly, Doctor access to patient-related information is associated with authorized appointment relationships.

The architecture also supports automated healthcare workflows. When a baby is registered, active vaccination schedule templates maintained by the Government module can be used to generate the applicable vaccination records automatically. Similarly, appointment requests submitted by Mothers are processed by checking the Doctor's configured availability and capacity before a booking is created. These workflows allow information entered through one module to be used by another module while maintaining the appropriate access restrictions.

Another important component of the architecture is the notification mechanism, which provides users with information about relevant system events. Notifications can be generated for important workflow changes such as appointment-related activities and can be delivered to the appropriate recipient. This allows the different users of the system to remain informed about activities that require their attention without providing access to unrelated information.

The architecture follows a modular design in which the four main modules operate as separate role-specific components while sharing the common Django backend and database. The Mother module handles beneficiary-related activities, the ASHA Worker module manages assigned cases, the Doctor module handles availability and appointments, and the Government module manages administrative resources and system-level information. Although these modules perform different functions, they interact through the common application and shared healthcare data.

The layered and modular architecture of MARGAM supports maintainability and future expansion of the system. Additional functionality can be incorporated into the application without requiring major changes to the existing structure. Overall, the architecture integrates the web interface, Django backend, role-based modules, Django ORM and SQLite database to provide a centralized platform for managing maternal and child

healthcare information and coordinating the activities of Mothers, ASHA Workers, Doctors and Government Officials.

5.4 Case Study

To evaluate the practical working of the MARGAM system, a typical maternal and child healthcare scenario can be considered. Suppose a mother registers in the MARGAM application and adds her baby to the system. The system connects the baby with the relevant maternal record and assigned ASHA Worker, allowing the healthcare information to be maintained in an organized manner.

When the baby is registered, the system uses the active vaccination schedule maintained by the Government Official to generate the applicable vaccination records for the baby. The mother and authorized healthcare personnel can view the vaccination information through their respective modules. As the baby grows, authorized users can record growth measurements, allowing the baby's growth information to be maintained as a sequence of historical records. If the mother requires a consultation with a Doctor, the Doctor configures available dates and appointment capacity through the Doctor module. The mother can then select an eligible appointment and submit a request, which is checked against the doctor's availability and capacity before being created. The Doctor can review and confirm or reject the request, while important appointment status changes can be communicated through notifications.

The assigned ASHA Worker can access the mother's permitted information and follow up on the assigned case. The ASHA Worker can review baby information, record relevant growth details, and update eligible vaccination information. The mother can also communicate with the assigned ASHA Worker through the available communication facility when assistance or follow-up is required. Access to these records is controlled according to the user's role and relationship with the respective healthcare information.

This case study demonstrates how MARGAM integrates maternal records, baby registration, vaccination management, growth tracking, appointment management, notifications, and role-based healthcare activities within a single platform. By connecting the activities of Mothers, ASHA Workers, Doctors, and Government Officials, the system provides a structured approach for coordinating maternal and child healthcare information and related workflows.

CHAPTER 6

IMPLEMENTATION DETAILS

6.1 Implementation Procedures

The implementation of MARGAM was carried out in a series of stages to ensure that the different components of the system could be developed and integrated in an organised manner. The process began with analysing the requirements of a maternal and child healthcare management platform and identifying the main functions required for Government Officials, ASHA Workers, Doctors and Mothers. Based on these requirements, the system structure, user roles, database requirements and major application modules were planned before development.

The frontend of the system was developed to provide a simple interface through which users can register, log in, manage healthcare information and access the services available to them. Separate functionalities were provided for Government Officials, ASHA Workers, Doctors and Mothers according to their respective roles. The backend was implemented using Django to manage authentication, maternal and child records, vaccination information, appointments, notifications and other application data. SQLite3 was used to store and retrieve the required information.

After the basic user and healthcare record management features were implemented, the role-based modules were integrated. The Government module was connected with the management of Doctors, ASHA Workers, vaccination schedules and notifications. The ASHA Worker module was integrated with assigned mothers, baby records, growth information and vaccination updates. The Doctor module was connected with availability management, appointment requests and patient-related information, while the Mother module was integrated with baby records, vaccination schedules, appointments and healthcare information.

The automatic vaccination feature was implemented to generate applicable vaccination records when a baby is registered. The appointment management feature was also implemented to check Doctor availability and process appointment requests. Communication functionality was integrated between Mothers and their assigned ASHA Workers to support follow-up and assistance. Relevant healthcare information

was stored in the database and made available to authorized users according to their roles.

Finally, the notification, dashboard and reporting features were integrated with the completed modules. Government Officials can monitor system information and manage notifications, while Mothers, ASHA Workers and Doctors can access the information relevant to their respective responsibilities. Each major module was tested during development, and integration testing was performed to verify that the different components worked together correctly. This step-by-step implementation process helped build MARGAM as a complete platform for managing maternal and child healthcare activities.

6.2 Testing Methods and Results

Testing was carried out throughout the development of MARGAM to verify that the different modules function correctly and work together as expected. The testing process used sample data representing the activities normally performed in the maternal and child healthcare system. Both valid and invalid inputs were included to check the system under different conditions.

The testing dataset was prepared specifically for the MARGAM application and consisted of sample records required for different modules. It included **user data** containing Government Official, ASHA Worker, Doctor and Mother accounts, **maternal and baby data** containing sample mother and baby records, **vaccination data** containing schedules, due dates and statuses, **appointment data** containing Doctor availability and appointment requests, and **growth, notification and communication data** for testing baby growth, notifications and communication between Mothers and ASHA Workers. These datasets were used only for testing and did not represent real patient information.

The main testing strategies included **unit testing, integration testing, functional testing and system testing**. Unit testing was used to verify individual functions such as authentication, user management, baby registration, vaccination generation and appointment booking. Integration testing checked whether different modules exchanged data correctly. Functional testing verified whether each feature produced the expected result, while system testing evaluated the complete application workflow.

Testing Results

Module / Feature	Test Performed	Result
User Authentication	Valid and invalid login details	Passed
User Registration	Mother registration and account creation	Passed
Government Management	Doctor, ASHA Worker and vaccination schedule management	Passed
Mother Management	Mother and baby information management	Passed
Vaccination Management	Vaccination generation and status updates	Passed
Growth Tracking	Adding and viewing baby growth records	Passed
Appointment Management	Doctor availability and appointment booking	Passed
Notifications	Sending and displaying notifications	Passed
ASHA Worker Communication	Communication between Mother and ASHA Worker	Passed
Dashboard	Display of role-specific information	Passed

The vaccination module was tested using the prepared vaccination dataset to verify automatic record generation, due dates, vaccination status updates and different baby registration conditions. The appointment feature was tested using different Doctor availability and booking conditions to verify successful appointment requests and appropriate handling of unavailable slots. The role-based access feature was also tested with different user accounts to verify that users could access only the functions permitted for their respective roles.

The healthcare record features were tested using separate Government Official, ASHA Worker, Doctor and Mother accounts. User authentication, dashboard access, baby registration, vaccination management, appointment processing and communication were checked under different conditions. The notification and reporting-related functions were tested by entering sample information and verifying that the required information was displayed to the appropriate users.

The testing results showed that the major functions of MARGAM worked according to the expected requirements. Issues identified during testing were corrected and the affected modules were tested again to confirm proper operation.

CHAPTER 7

ADVANTAGES AND DISADVANTAGES

7.1 Advantages

MARGAM provides several advantages by bringing the major activities of maternal and child healthcare management into a single platform. It reduces the need to maintain healthcare information through separate systems and provides an organised environment for Government Officials, ASHA Workers, Doctors and Mothers.

- **Unified Healthcare Platform:** Maternal records, baby information, vaccination, growth tracking, appointments and notifications are managed within the same platform.
- **Role-Based Access:** Different users are provided with appropriate functions according to their roles, helping maintain organized and controlled access to healthcare information.
- **Automatic Vaccination Management:** Vaccination records can be generated based on the defined vaccination schedule, reducing manual entry and helping maintain a structured vaccination history.
- **Improved Healthcare Coordination:** Government Officials, ASHA Workers, Doctors and Mothers can access relevant information through their respective modules, improving coordination between stakeholders.
- **Baby Growth Monitoring:** Growth information can be recorded and viewed over time, helping users monitor the development of babies.
- **Efficient Appointment Management:** Mothers can request appointments with Doctors, while Doctors can manage their availability and appointment requests through the system.
- **Centralised Records:** Maternal, baby, vaccination, growth and appointment information can be maintained in a centralized database.
- **Notifications and Alerts:** The system can provide relevant notifications and vaccination alerts to help users remain aware of important healthcare activities.
- **Direct Communication:** Mothers can communicate with their assigned ASHA Workers through the system, supporting follow-up and healthcare assistance.

- **Convenient Web-Based Access:** Since MARGAM is a web-based application, users can access the system through a supported web browser without requiring a separate application.

Overall, these advantages make MARGAM a practical platform for managing maternal and child healthcare activities in an organised manner. By combining healthcare records, vaccination management, growth tracking, appointments, notifications and communication, the system can simplify the management of different healthcare activities and improve coordination among the users.

7.2 Disadvantages

Although MARGAM provides an integrated environment for maternal and child healthcare management, the system has some limitations. These limitations are mainly related to internet connectivity, data security, system resources and the scope of the current implementation.

- **Internet Dependency:** Stable internet connectivity is required to access the web application and use its different functions effectively.
- **Device Requirements:** Users require a suitable device and supported web browser to access the system and its features.
- **Data Security Requirements:** Since the system handles maternal and child healthcare information, appropriate security measures are required to protect sensitive information from unauthorized access.
- **Database Limitations:** SQLite3 is suitable for the current project and small-scale usage, but a more advanced database system may be required when the number of users and records increases significantly.
- **Limited Offline Support:** The current web application mainly depends on an active internet connection, which may limit access in areas with poor or unavailable connectivity.
- **Manual Data Accuracy:** The accuracy of healthcare information depends on the correctness of the data entered by authorized users.
- **Notification Limitations:** The effectiveness of healthcare notifications depends on the availability and proper use of the notification features within the system.

- **Scalability Requirements:** As the number of users and healthcare records increases, additional server and database resources may be required to maintain system performance.
- **Initial Learning Requirement:** Users may require some time to become familiar with the different modules and functions of the system before using it effectively.

Overall, these limitations do not prevent the use of MARGAM, but they indicate areas that require attention during deployment and future development. Improving security, scalability, offline capabilities, database performance and notification mechanisms can further strengthen the system.

CHAPTER 8

RESULT AND DISCUSSION

8.1 Analysis

The analysis of MARGAM was carried out based on the implementation and testing of its major modules. The results indicate that the developed system can support the main activities involved in maternal and child healthcare management within a single web-based environment. The core functions, including user authentication, healthcare record management, vaccination management, baby growth tracking, appointment management, notifications and communication, were tested using suitable sample data and different input conditions.

The testing showed that Government Officials, ASHA Workers, Doctors and Mothers could access the system according to their assigned roles and use the available functions successfully. The Government module supported management of Doctors, ASHA Workers, vaccination schedules and notifications, while the ASHA Worker and Mother modules supported the management and viewing of relevant maternal and child healthcare information.

The analysis also showed that the automatic vaccination feature was able to generate applicable vaccination records when a baby was registered. The appointment module allowed Mothers to request appointments based on Doctor availability, while Doctors could review and manage the requests. The growth tracking feature allowed baby growth information to be recorded and viewed over time.

Another important observation was the coordination between the different modules during the complete healthcare workflow. Information entered or generated at one stage could be used by related features in later stages, such as using baby details for vaccination record generation and appointment information for Doctor and Mother management. This integration helped maintain healthcare information in a structured manner and reduced unnecessary duplication of information.

Overall, the results demonstrate that the major requirements identified for MARGAM were addressed by the implemented system. At the same time, the analysis highlights certain practical limitations, such as dependence on stable internet connectivity, database capacity and proper data entry. These factors can influence the performance and

effectiveness of some features during actual use. The results therefore indicate that MARGAM provides a functional foundation for managing maternal and child healthcare while also identifying areas that can be improved in future versions.

The key observations obtained from the analysis of the system include:

- **Successful Role Management:** Government Officials, ASHA Workers, Doctors and Mothers were able to access the functions available to their respective roles.
- **Integrated Healthcare Management:** The major healthcare activities could be performed within the same platform, reducing the need to use multiple separate systems.
- **Effective Record Management:** Maternal, baby, vaccination and growth information could be stored and accessed by authorized users.
- **Automatic Vaccination Generation:** Applicable vaccination records were generated based on the baby's registration details and active vaccination schedule.
- **Appointment Management:** Mothers could request appointments while Doctors could manage their availability and appointment requests.
- **Growth Tracking:** Baby growth information could be recorded and viewed to support monitoring of development.
- **Notification Support:** Relevant notifications could be created and provided to the appropriate users.
- **Communication:** Mothers and assigned ASHA Workers could communicate through the system for follow-up and assistance.
- **Module Integration:** The tested modules worked together to support the overall healthcare management workflow.
- **Identified Limitations:** The analysis also identified practical dependencies such as stable internet connectivity, database capacity and accurate data entry.

Overall, the observations indicate that MARGAM achieved the major functional requirements defined for the proposed maternal and child healthcare management platform.

CHAPTER 9

CONCLUSION

9.1 Summary

MARGAM is a web-based maternal and child healthcare management platform developed to simplify and organise the management of healthcare activities. The system brings important activities such as maternal record management, baby registration, vaccination management, growth tracking, appointment management, notifications and communication into a single platform. This reduces the need to depend on separate methods for maintaining and managing healthcare information.

The platform provides separate roles for Government Officials, ASHA Workers, Doctors and Mothers. Government Officials can manage system-level resources, vaccination schedules, notifications and reports. ASHA Workers can manage assigned mothers and babies and provide follow-up support. Doctors can manage availability and appointments, while Mothers can manage their information, access baby-related records, view vaccination schedules and request appointments.

MARGAM also includes automatic vaccination-record generation, which helps create applicable vaccination records when a baby is registered. The system supports growth monitoring, appointment management and communication between Mothers and ASHA Workers. These features help maintain relevant healthcare information in an organised manner and improve coordination between the different users.

The development and testing of the system showed that the major modules can work together as part of a complete maternal and child healthcare management workflow. Overall, MARGAM provides a structured approach for managing maternal and child healthcare information and can reduce the manual effort involved in maintaining and coordinating healthcare activities.

9.2 Limitations

Although MARGAM provides the required features for managing maternal and child healthcare activities, the system has some practical limitations.

- **Internet Dependency:** Access to the web application and its major features requires a stable internet connection.

- **Device Requirements:** The quality of system usage may depend on the user's device, browser and available hardware.
- **Data Security Requirements:** Healthcare information requires appropriate security measures to protect maternal and child records from unauthorized access.
- **Database Limitations:** The current SQLite3 database is suitable for the project, but a more scalable database may be required when the number of users and records increases.
- **Limited Offline Support:** The current implementation mainly depends on an active internet connection and may therefore be difficult to access in areas with poor connectivity.
- **Data Entry Dependency:** The accuracy of healthcare information depends on the correctness of the information entered by authorized users.
- **Notification Limitations:** The effectiveness of notifications depends on the availability and proper functioning of the notification features.
- **Performance Issues:** A large number of simultaneous users may require additional server resources and performance optimisation.
- **Initial Learning Requirement:** New users may need some time to become familiar with the different features of the platform.

These limitations provide areas for further improvement and optimisation as the system is developed into a larger-scale healthcare platform.

9.3 Future Scope

MARGAM can be further improved by adding advanced features and improving the existing modules. Future development can focus on scalability, security, automation, accessibility and better healthcare support.

Possible future enhancements include:

- **Scalable Infrastructure:** Deploy the system using scalable cloud infrastructure to support a larger number of users and healthcare records.
- **Improved Offline Support:** Introduce stronger offline data-entry and synchronization facilities for areas with limited internet connectivity.
- **Mobile Support:** Develop dedicated mobile support for Mothers, ASHA Workers and other healthcare users.

- **Improved Security:** Add stronger authentication, access control and security mechanisms for protecting healthcare information.
- **Advanced Reporting:** Provide more detailed and customizable reports for Government Officials and healthcare administrators.
- **Automated Health Monitoring:** Introduce additional automated mechanisms for identifying important vaccination, growth and healthcare follow-up requirements.

These improvements can help MARGAM evolve from a project-level healthcare platform into a more comprehensive system for maternal and child healthcare management.

9.4 Applications

MARGAM can be applied in different situations where maternal and child healthcare activities need to be managed digitally. Its combination of healthcare records, vaccination, growth monitoring, appointments and communication makes it suitable for several use cases.

- **Maternal Healthcare Management:** Healthcare organisations can use the platform to maintain and manage maternal healthcare information.
- **Child Healthcare Monitoring:** The system can be used to maintain baby records, vaccination information and growth details.
- **Vaccination Management:** Healthcare workers can use the system to manage vaccination schedules and monitor vaccination status.
- **ASHA Worker Support:** ASHA Workers can use the platform to manage assigned mothers and babies and provide follow-up assistance.
- **Doctor Appointment Management:** Doctors and Mothers can use the system to manage availability and appointment requests.
- **Government Healthcare Monitoring:** Government Officials can use the platform to manage healthcare resources, vaccination schedules, notifications and reports.

Overall, MARGAM demonstrates how web technologies can be combined to create a structured and convenient environment for maternal and child healthcare management. The project provides a foundation that can be extended with stronger security, improved analytics, enhanced offline capabilities and scalable infrastructure in future versions

CHAPTER 10

APPENDICES

10.1 Sample Code

```

margam> core> models.py
1 from django.db import models
2 from django.conf import settings
3 from django.utils import timezone
4
5
6
7 class TimeStampedModel(models.Model):
8     """Abstract base - every model in margam gets timestamps"""
9     created_at = models.DateTimeField(auto_now_add=True)
10    updated_at = models.DateTimeField(auto_now=True)
11
12    class Meta:
13        abstract = True
14
15
16 class Panchayat(TimeStampedModel):
17     name = models.CharField(max_length=100, unique=True)
18     district = models.CharField(max_length=100)
19     block = models.CharField(max_length=100, blank=True)
20
21     class Meta:
22         ordering = ['district', 'name']
23         verbose_name_plural = 'Panchayats'
24
25     def __str__(self):
26         return f'{self.name}, {self.district}'
27
28
29 class Disease(TimeStampedModel):
30     name = models.CharField(max_length=150, unique=True)
31     description = models.TextField()
32     symptoms = models.TextField(blank=True)
33     prevention = models.TextField(blank=True)
34     added_by = models.ForeignKey(
35         settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
36         null=True, related_name='diseases_added')
37     is_active = models.BooleanField(default=True)
38
39     class Meta:
40         ordering = ['name']
41
42     def __str__(self):
43         return self.name
44
45
46 class HealthTip(TimeStampedModel):
47     class Category(models.TextChoices):
48         NUTRITION = 'nutrition', 'Nutrition'
49         HYGIENE = 'hygiene', 'Hygiene'
50         PRENATAL = 'prenatal', 'Prenatal'
51         POSTNATAL = 'postnatal', 'Postnatal'
52         INFANT_CARE = 'infant_care', 'Infant Care'
53         VACCINATION = 'vaccination', 'Vaccination'
54         GENERAL = 'general', 'General'
55
56     title = models.CharField(max_length=200)
57     content = models.TextField()
58     category = models.CharField(max_length=20, choices=Category.choices,
59                                blank=True)
60     added_by = models.ForeignKey(
61         settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
62         null=True, related_name='tips_added')
63     is_active = models.BooleanField(default=True)
64
65     class Meta:
66         ordering = ['created_at']
67
68     def __str__(self):
69         return self.title
70
71
72 class Food(TimeStampedModel):
73     class TargetGroup(models.TextChoices):
74         PREGNANT = 'pregnant', 'Pregnant Mother'
75         LACTATING = 'lactating', 'Lactating Mother'
76         INFANT = 'infant', 'Infant (0-5 months)'
77         TODDLER = 'toddler', 'Toddler (6-24 months)'
78         GENERAL = 'general', 'General'
79
80     name = models.CharField(max_length=150)
81     description = models.TextField()
82     nutrients = models.TextField(blank=True)
83     benefits = models.TextField(blank=True)
84     target_group = models.CharField(max_length=20, choices=TargetGroup.choices,
85                                    blank=True)
86     added_by = models.ForeignKey(
87         settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
88         null=True, related_name='foods_added')
89     is_active = models.BooleanField(default=True)
90
91     class Meta:
92         ordering = ['target_group', 'name']
93
94     def __str__(self):
95         return f'{self.name} ({self.get_target_group_display()})'
96
97
98 class VaccinationScheduleTemplate(TimeStampedModel):
99     """
100     Government defined standard schedule.
101     Each row = one dose at a specific age.
102     Mother's actual records reference this.
103     """
104     vaccine_name = models.CharField(max_length=150)
105     age_in_weeks = models.PositiveIntegerField(help_text='Age at which to give vaccine')
106     dose_number = models.PositiveIntegerField(default=1)
107     description = models.TextField(blank=True)
108     is_mandatory = models.BooleanField(default=True)
109     added_by = models.ForeignKey(
110         settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
111         null=True, related_name='schedules_added')
112     is_active = models.BooleanField(default=True)
113
114     class Meta:
115         ordering = ['age_in_weeks', 'dose_number']
116         unique_together = ('vaccine_name', 'age_in_weeks', 'dose_number')
117         verbose_name = 'Vaccination Schedule'
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Sample Code for Core Healthcare Models

```

margam> mother> models.py
1 from django.db import models
2 from django.conf import settings
3 from django.utils import timezone
4 from core.models import TimeStampedModel, VaccinationScheduleTemplate
5
6
7 class MotherProfile(TimeStampedModel):
8     class PregnancyStatus(models.TextChoices):
9         PREGNANT = 'pregnant', 'Pregnant'
10        POSTNATAL = 'postnatal', 'Postnatal'
11        LACTATING = 'lactating', 'Lactating'
12        OTHER = 'other', 'Other'
13
14    user = models.OneToOneField(
15        settings.AUTH_USER_MODEL, on_delete=models.CASCADE, related_name='mother_profile')
16    asha_worker = models.ForeignKey(
17        'asha.AshaWorkerProfile', on_delete=models.SET_NULL,
18        null=True, blank=True, related_name='mothers')
19    date_of_birth = models.DateField(null=True, blank=True)
20    address = models.TextField(blank=True)
21    guardian_name = models.CharField(max_length=100, blank=True)
22    guardian_phone = models.CharField(max_length=15, blank=True)
23    pregnancy_status = models.CharField(max_length=15, choices=PregnancyStatus.choices, blank=True)
24    expected_due_date = models.DateField(null=True, blank=True)
25    blood_group = models.CharField(max_length=5, blank=True)
26    notes = models.TextField(blank=True)
27    is_active = models.BooleanField(default=True)
28
29    class Meta:
30        verbose_name = 'Mother Profile'
31
32    def __str__(self):
33        return self.user.get_full_name() or self.user.username
34
35
36
37
38 class BabyProfile(TimeStampedModel):
39     class Gender(models.TextChoices):
40         MALE = 'M', 'Male'
41         FEMALE = 'F', 'Female'
42
43    mother = models.ForeignKey(MotherProfile, on_delete=models.CASCADE, related_name='babies')
44    name = models.CharField(max_length=100)
45    date_of_birth = models.DateField()
46    gender = models.CharField(max_length=1, choices=Gender.choices)
47    birth_weight = models.DecimalField(max_digits=4, decimal_places=2, null=True, blank=True)
48    blood_group = models.CharField(max_length=5, blank=True)
49    notes = models.TextField(blank=True)
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models.py
margam > mother > models.py > MotherProfile > PregnancyStatus
90 class VaccinationRecord(TimeStampedModel):
91     schedule = models.ForeignKey(VaccinationScheduleTemplate, on_delete=models.CASCADE)
92     due_date = models.DateField(db_index=True)
93     given_date = models.DateField(null=True, blank=True)
94     given_by = models.ForeignKey(
95         settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
96         null=True, blank=True, related_name='vaccinations_administered'
97     )
98     status = models.CharField(max_length=10, choices=Status.choices, default='PENDING')
99     batch_number = models.CharField(max_length=50, blank=True)
100     notes = models.TextField(blank=True)
101     alert_sent = models.BooleanField(default=False)
102
103     class Meta:
104         ordering = ['-due_date']
105
106     def is_overdue(self):
107         return self.status == 'pending' and self.due_date < timezone.now().date()
108
109     def __str__(self):
110         return f'{self.baby.name} - {self.schedule} [{self.get_status_display()}]'
111
112 class Booking(TimeStampedModel):
113     class Status(models.TextChoices):
114         PENDING = 'pending', 'Pending'
115         CONFIRMED = 'confirmed', 'Confirmed'
116         REJECTED = 'rejected', 'Rejected'
117         COMPLETED = 'completed', 'Completed'
118         CANCELLED = 'cancelled', 'Cancelled'
119
120     mother = models.ForeignKey(MotherProfile, on_delete=models.CASCADE, related_name='bookings')
121     doctor = models.ForeignKey('doctor.DoctorProfile', on_delete=models.CASCADE, related_name='bookings')
122     appointment_date = models.DateField(db_index=True)
123     reason = models.TextField(blank=True)
124     status = models.CharField(max_length=10, choices=Status.choices, default='PENDING')
125     doctor_notes = models.TextField(blank=True)
126     rejection_reason = models.TextField(blank=True)
127
128     class Meta:
129         ordering = ['-appointment_date']
130
131     def __str__(self):
132         return f'{self.mother} on {self.appointment_date} [{self.get_status_display()}]'
133
134 class AssistanceRequest(TimeStampedModel):
135     class Category(models.TextChoices):
136         ASHA_UNREACHABLE = 'asha_unreachable', 'ASHA Worker Unreachable'
137
138     mother = models.ForeignKey(MotherProfile, on_delete=models.CASCADE, related_name='assistance_requests')
139     category = models.CharField(max_length=30, choices=Category.choices, default='ASHA_UNREACHABLE')
140     message = models.TextField()
141     status = models.CharField(max_length=15, choices=Status.choices, default='PENDING')
142     resolution_notes = models.TextField(blank=True)
143
144     class Meta:
145         ordering = ['-created_at']
146         verbose_name = 'Assistance Request'
147
148     def __str__(self):
149         return f'Assistance for {self.mother} [{self.get_status_display()}]'

```

Sample Code for Mother and Baby Models

```

signals.py
margam > core > signals.py > ...
1 from django.db.models.signals import post_save
2 from django.dispatch import receiver
3 from datetime import timedelta
4
5
6 @receiver(post_save, sender='mother.BabyProfile')
7 def generate_vaccination_records(sender, instance, created, **kwargs):
8     if created:
9         from core.models import VaccinationScheduleTemplate
10        from mother.models import VaccinationRecord
11        templates = VaccinationScheduleTemplate.objects.filter(
12            is_active=True
13        )
14        records = []
15        for template in templates:
16            due_date = (instance.date_of_birth +
17                timedelta(weeks=template.age_in_weeks))
18            records.append(VaccinationRecord(
19                baby=instance,
20                schedule=template,
21                due_date=due_date,
22                status='pending'
23            ))
24        VaccinationRecord.objects.bulk_create(records)

```

Sample Code for Automatic Vaccination Generation

```

1 from functools import wraps
2 from django.shortcuts import redirect
3 from django.contrib import messages
4
5
6 def role_required(*roles):
7     """
8     Usage:
9     @role_required('government')
10    @role_required('doctor', 'government')
11    """
12     def decorator(view_func):
13         @wraps(view_func)
14         def wrapper(request, *args, **kwargs):
15             if not request.user.is_authenticated:
16                 return redirect('accounts:login')
17             if request.user.role not in roles:
18                 messages.error(request, 'You are not authorised to access this page.')
19                 return redirect(request.user.get_dashboard_url())
20             return view_func(request, *args, **kwargs)
21         return wrapper
22     return decorator
23
24 # Convenience aliases
25 government_required = role_required('government')
26 doctor_required = role_required('doctor')
27 mother_required = role_required('mother')
28
29 def asha_required(view_func):
30     """
31     Ensures user is an authenticated ASHA worker AND has been approved by Government.
32     Unapproved ASHA workers are redirected to the verification status page.
33     """
34     @wraps(view_func)
35     def wrapper(request, *args, **kwargs):
36         if not request.user.is_authenticated:
37             return redirect('accounts:login')
38         if not request.user.is_asha_worker:
39             messages.error(request, 'You are not authorised to access this page.')
40             return redirect(request.user.get_dashboard_url())
41         profile = getattr(request.user, 'asha_profile', None)
42         if profile and not profile.is_approved:
43             return redirect('asha:verification_status')
44         return view_func(request, *args, **kwargs)
45     return wrapper
46
47 def doctor_required(view_func):
48     """
49     Ensures user is an authenticated doctor AND has been approved by Government.
50     Unapproved doctors are redirected to the verification status page.
51     """
52     @wraps(view_func)
53     def wrapper(request, *args, **kwargs):
54         if not request.user.is_authenticated:
55             return redirect('accounts:login')
56         if not request.user.is_doctor:
57             messages.error(request, 'You are not authorised to access this page.')
58             return redirect(request.user.get_dashboard_url())
59         profile = getattr(request.user, 'doctor_profile', None)
60         if profile and not profile.is_approved:
61             return redirect('doctor:verification_status')
62         return view_func(request, *args, **kwargs)
63     return wrapper
64
65

```

Sample Code for Role Based Access Control

```

1 from django import forms
2 from django.contrib import messages
3 from django.shortcuts import redirect, render
4 from django.utils import timezone
5
6 from core.decorators import mother_required
7 from core.models import Disease, Food, HealthInfo, Message, Notification
8 from mother.models import DoctorAvailability, DoctorProfile, Booking, MotherProfile, VaccinationRecord
9
10 @mother_required
11 def dashboard(request):
12     mother = request.user.mother_profile
13     today = timezone.now().date()
14     # Calculate pregnancy progress if pregnant and has expected due date
15     pregnancy_info = None
16     if mother.pregnancy_status == MotherProfile.PREGNANT:
17         due_date = mother.expected_due_date
18         days_remaining = (due_date - today).days
19         # Standard gestation period is 280 days
20         gestational_days = 280 - days_remaining
21         if gestational_days < 0:
22             gestational_days = 0
23         weeks = gestational_days // 7
24         days = gestational_days % 7
25         progress_pct = int((gestational_days / 280.0) * 100)
26
27     if gestational_days < 91:
28         trimester = 1
29         trimester_name = "1st Trimester"
30         trimester_desc = "Prenatal Phase"
31     elif gestational_days < 182:
32         trimester = 2
33         trimester_name = "2nd Trimester"
34         trimester_desc = "Growth & Movement"
35     else:
36         trimester = 3
37         trimester_name = "3rd Trimester"
38         trimester_desc = "Preparation & Delivery"
39
40     pregnancy_info = {
41         'weeks': weeks,
42         'days': days,
43         'days_remaining': max(0, days_remaining),
44         'progress_pct': progress_pct,
45         'trimester': trimester,
46         'trimester_name': trimester_name,
47         'trimester_desc': trimester_desc,
48     }
49
50 # Get babies data with latest growth metrics and vaccination progress
51 babies_data = []
52 for baby in babies:
53     latest_growth = baby.growth_records.order_by('-record_date').first()
54     total_vax = baby.vaccinations.count()
55     given_vax = baby.vaccinations.filter(status='VaccinationRecord.STATUS_PENDING').count()
56     vax_progress = int((given_vax / total_vax * 100)) if total_vax > 0 else 0
57
58     babies_data.append({
59         'baby': baby,
60         'latest_growth': latest_growth,
61         'total_vax': total_vax,
62         'given_vax': given_vax,
63         'vax_progress': vax_progress,
64     })
65
66 # Get upcoming vaccinations (up to 3)
67 upcoming_vaccinations = VaccinationRecord.objects.filter(
68     baby__mother=mother,
69     status='VaccinationRecord.STATUS_PENDING',
70     due_date__gte=today,
71 ).select_related('baby', 'schedule').order_by('due_date')[:3]
72
73 # Get upcoming bookings (future bookings that are pending or confirmed)
74 upcoming_bookings = Booking.objects.filter(
75     mother=mother,
76     appointment_date__gte=today,
77     status__in=[Booking.STATUS_PENDING, Booking.STATUS_CONFIRMED],
78 ).select_related('doctor_user').order_by('appointment_date')
79
80 # Next vaccine alert (first upcoming pending vaccine)
81 next_vaccine = upcoming_vaccinations.first() if upcoming_vaccinations.exists() else None
82
83 # Pending booking (for backward compatibility / quick actions alert)
84 pending_booking = Booking.objects.filter(mother=mother, status=Booking.STATUS_PENDING).first()
85
86 # Total pending vaccinations count
87 pending_vaccines_count = VaccinationRecord.objects.filter(
88     baby__mother=mother,
89     status='VaccinationRecord.STATUS_PENDING').count()
90
91 # Unread alerts count
92 unread_alerts = request.user.received_notifications.filter(is_read=False).count()
93
94 # Pending assistance - AssistanceRequest objects filter
95 pending_assistance = AssistanceRequest.objects.filter(
96     mother=mother, status='AssistanceRequest.STATUS_PENDING').first()
97
98 return render(request, 'mother/dashboard.html', {
99     'mother': mother,
100     'babies_data': babies_data,
101     'pregnancy_info': pregnancy_info,
102     'next_vaccine': next_vaccine,
103     'upcoming_vaccinations': upcoming_vaccinations,
104     'upcoming_bookings': upcoming_bookings,
105     'pending_vaccines_count': pending_vaccines_count,
106     'pending_assistance': pending_assistance,
107     'unread_alerts': unread_alerts,
108     'pending_booking': pending_booking,
109 })

```

```

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MARGAM - experiment

# views.py
margin > mother > views.py --
149 def book_confirm(request):
150     der_book_confirm(request)
151     appointment_date=apt_date,
152     reason=reason,
153     status=status.PENDING,
154     )
155     messages.success(request, 'Appointment booked! Waiting for doctor')
156     return redirect('motherbookings')
157     return redirect('motherbook_doctor')
158
159 @mother_required
160 def bookings(request):
161     mother = request.user.mother_profile
162     bookings_qs = Booking.objects.filter(mother=mother).select_related('appointment')
163     return render(request, 'motherbookings.html', {'bookings': bookings_qs,
164
165 @mother_required
166 def booking_detail(request, pk):
167     mother = request.user.mother_profile
168     booking = get_object_or_404(Booking, pk=pk, mother=mother)
169     return render(request, 'motherbooking_detail.html', {'booking': booking,
170
171 @mother_required
172 def booking_cancel(request, pk):
173     if request.method == 'POST':
174         return redirect('motherbookings')
175     mother = request.user.mother_profile
176     booking = get_object_or_404(Booking, pk=pk, mother=mother, status='PENDING')
177     booking.status = Booking.Status.CANCELLED
178     booking.save(update_fields=('status',))
179     messages.success(request, 'Booking cancelled.')
180     return redirect('motherbookings')
181
182 @mother_required
183 def baby_health(request):
184     mother = request.user.mother_profile
185     babies = mother.babies.all()
186     return render(request, 'mother_baby_health.html', {'babies': babies,
187
188 @mother_required
189 def baby_add(request):
190     class BabyForm(dj_forms.ModelForm):
191         model = BabyProfile
192         fields = ('name', 'date_of_birth', 'gender', 'birth_weight',
193                 'date_of_birth')
194         dj_forms.DateInput(attrs={'type': 'date'})
195
196 class BabyForm(dj_forms.ModelForm):
197     model = BabyProfile
198     fields = ('name', 'date_of_birth', 'gender', 'birth_weight',
199             'date_of_birth')
200     dj_forms.DateInput(attrs={'type': 'date'})
201
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211             'date_of_birth')
212     dj_forms.DateInput(attrs={'type': 'date'})
213
202 ef baby_detail(request):
203     class BabyForm(dj_forms.ModelForm):
204         model = BabyProfile
205         fields = ('name', 'date_of_birth', 'gender', 'birth_weight',
206                 'date_of_birth')
207         dj_forms.DateInput(attrs={'type': 'date'})
208     if request.method == 'POST':
209         form = BabyForm(request.POST)
210         if form.is_valid():
211             baby = form.save(commit=False)
212             baby.mother = request.user.mother_profile
213             baby.save()
214             messages.success(request, f'{baby.name} added! Vaccination records added!')
215             return redirect('mother_baby_detail', pk=baby.pk)
216         else:
217             form = BabyForm()
218     return render(request, 'mother/baby_add.html', {'form': form})
219
220 mother_required
221 def baby_detail(request, pk):
222     mother = request.user.mother_profile
223     baby = get_object_or_404(BabyProfile, pk=pk, mother=mother)
224     growth_records = baby.growth_records.order_by('record_date')
225     vaccinations_qs = baby.vaccinations.select_related('schedule').order_by('date')
226     import json
227     dates = [str(record.date) for r in growth_records]
228     weights = [float(r.weight_kg) if r.weight_kg else None for r in growth_records]
229     heights = [float(r.height_cm) if r.height_cm else None for r in growth_records]
230     return render(request, 'mother/baby_detail.html', {'baby': baby,
231         'growth_records': growth_records,
232         'vaccinations': vaccinations_qs,
233         'dates_json': json.dumps(dates),
234         'weights_json': json.dumps(weights),
235         'heights_json': json.dumps(heights),
236     })
237
238 @mother_required
239 def growth_add(request, baby_pk):
240     mother = request.user.mother_profile
241     baby = get_object_or_404(BabyProfile, pk=baby_pk, mother=mother)
242
243     class GrowthForm(dj_forms.ModelForm):
244         model = BabyGrowthRecord
245         fields = ('record_date', 'weight_kg', 'height_cm', 'head_circumference',
246                 'record_date')
247         dj_forms.DateInput(attrs={'type': 'date'})
248     class Meta:
249         model = BabyGrowthRecord
250         fields = ('record_date', 'weight_kg', 'height_cm', 'head_circumference',
251                 'record_date')
252         dj_forms.DateInput(attrs={'type': 'date'})
253     if request.method == 'POST':
254         form = GrowthForm(request.POST)
255         if form.is_valid():
256             record = form.save(commit=False)
257             record.baby = baby
258             record.mother = mother
259             record.save()
260             messages.success(request, f'Growth record saved.')
261             return redirect('mother_baby_detail', pk=baby.pk)
262         else:
263             form = GrowthForm()
264     return render(request, 'mother/growth_add.html', {'form': form,
265         'baby': baby,
266
267 @mother_required
268 def vaccinations(request):
269     mother = request.user.mother_profile
270     today = timezone.now().date()
271     vaccinations_qs = VaccinationRecord.objects.filter(baby__mother=mother, date__lte=today)
272     return render(request, 'mother/vaccinations.html', {'vaccinations': vaccinations_qs,
273         'today': today,
274
275 @mother_required
276 def health_tips(request):
277     tips = HealthTip.objects.filter(is_active=True)
278     request.session['category'] = ''
279     if category:
280         tips = tips.filter(category=category)
281         categories = [(value, label) for r in HealthTip.objects.filter(is_active=True)]
282         return render(request, 'mother/health_tips.html', {'tips': tips,
283             'categories': categories,
284             'active_category': category,
285
286 @mother_required
287 def food(request):
288     mother = request.user.mother_profile
289     foods = Food.objects.filter(is_active=True)
290     status = mother.pregnancy_status
291     if status in ('pregnant', 'postnatal', 'lactating'):
292         foods = foods.filter(target_group__in=['status', 'general'])
293     return render(request, 'mother/food.html', {'foods': foods,
294
295 @mother_required
296 def diseases(request):
297     mother = request.user.mother_profile
298     diseases = Disease.objects.filter(is_active=True)
299     return render(request, 'mother/diseases.html', {'diseases': diseases,

```

```

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MARGAM - experiment

# views.py
margam > mother > # views.py >
345 def diseases(request):
346     diseases_qs = Disease.objects.filter(is_active=True)
347     return render(request, 'mother/diseases.html', {
348         'diseases': diseases_qs,
349     })
350
351 # mother_required
352 def chat(request):
353     mother = request.user.mother_profile
354     asha_worker = mother.asha_worker
355     messages_qs = None
356     if asha_worker:
357         if request.method == 'POST':
358             content = request.POST.get('content', '').strip()
359             if content:
360                 Message.objects.create(
361                     sender=request.user,
362                     recipient=asha_worker.user,
363                     content=content,
364                 )
365                 return redirect('mother:chat')
366             messages_qs = Message.objects.filter(
367                 sender__in=request.user, asha_worker=user,
368                 recipient__in=request.user, asha_worker=user,
369             ).order_by('created_at')
370             messages_qs = Message.objects.filter(
371                 sender=asha_worker.user,
372                 recipient=request.user,
373                 is_read=False,
374             ).update(is_read=True)
375             pending_assistance = AssistanceRequest.objects.filter(
376                 mother=mother, status=AssistanceRequest.Status.PENDING,
377             ).first()
378             return render(request, 'mother/chat.html', {
379                 'asha_worker': asha_worker,
380                 'messages': messages_qs,
381                 'pending_assistance': pending_assistance,
382             })
383
384 # mother_required
385 def chat_send(request):
386     return redirect('mother:chat')
387
388 # mother_required
389 def chat_messages_api(request):
390     from django.http import JsonResponse
391     mother = request.user.mother_profile
392     asha = mother.asha_worker
393     if not asha:
394         return JsonResponse({'messages': []})
395     after_id = int(request.GET.get('after', 0))
396     msgs = Message.objects.filter(
397         sender__in=request.user, asha=user,
398         recipient__in=request.user, asha=user,
399         id__gt=after_id,
400     )
401
402 # views.py
403
404 # chat_messages_api(request):
405 393     if chat_messages_api(request):
406         id__after_id,
407         .order_by('created_at')
408         data = [
409             'id': m.id,
410             'content': m.content,
411             'is_asha': m.sender == request.user,
412             'time': m.created_at.strftime('%H:%M'),
413         ] for m in msgs
414     return JsonResponse({'messages': data})
415
416 # mother_required
417 def alerts(request):
418     notifications = request.user.received_notifications.order_by('-created_at')
419     notifications.filter(is_read=False).update(is_read=True)
420     return render(request, 'mother/alerts.html', {'notifications': notifications})
421
422 # mother_required
423 def alert_read(request, pk):
424     if request.method == 'POST':
425         return redirect('mother:alerts')
426     notif = get_object_or_404(Notification, pk=pk, recipient=request.user)
427     notif.mark_read()
428     return redirect('mother:alerts')
429
430 # mother_required
431 def assistance_request(request):
432     mother = request.user.mother_profile
433     pending_req = AssistanceRequest.objects.filter(
434         mother=mother, status=AssistanceRequest.Status.PENDING,
435     ).first()
436     recent_requests = AssistanceRequest.objects.filter(mother=mother).or
437
438 if request.method == 'POST':
439     if pending_req:
440         messages.warning(request, 'You already have an assistance request')
441         return redirect('mother:dashboard')
442     category = request.POST.get('category', AssistanceRequest.Category.cholera)
443     message_text = request.POST.get('message', '').strip()
444     if not message_text:
445         messages.error(request, 'Please describe what assistance you need')
446         return render(request, 'mother/assistance_request.html', {
447             'mother': mother,
448             'pending_req': pending_req,
449             'categories': AssistanceRequest.Category.choices,
450             'recent_requests': recent_requests,
451         })
452     req = AssistanceRequest.objects.create(
453         mother=mother,
454         category=category,
455         message=message_text,
456         status=AssistanceRequest.Status.PENDING,
457     )
458
459 # views.py
460
461 # assistance_request(request):
462 459     def assistance_request(request):
463         status=AssistanceRequest.Status.PENDING,
464         )
465         from accounts.models import CustomUser
466         gov_users = CustomUser.objects.filter(role=CustomUser.Role.GOV)
467         notifs = [
468             Notification(
469                 sender=request.user,
470                 recipient=gov_user,
471                 title=f'Assistance Request: {mother.user.get_full_name()}',
472                 message=f'Mother {mother.user.get_full_name()} or mother {mother.user.get_full_name()} notified you of a {gov_user.get_full_name()} request for assistance'
473             ) for gov_user in gov_users
474         ]
475         if notifs:
476             Notification.objects.bulk_create(notifs)
477         Messages.success(request, 'Your assistance request has been submitted')
478         return redirect('mother:dashboard')
479
480 return render(request, 'mother/assistance_request.html', {
481     'mother': mother,
482     'pending_req': pending_req,
483     'categories': AssistanceRequest.Category.choices,
484     'recent_requests': recent_requests,
485 })
486

```

Sample Code for Mother Dashboard

Hosted Project Link :

Project Repository: <https://github.com/tonystark47477/MargamMiniProject.git>

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MELBIN K MADHU

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